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Transient, deadline-aware RAN slicing for NR-V2X using finite-capacity time-varying slice queues

Nurgüneş Yordanov *, Bülent Çavuşoğlu

Atatürk University, Department of Electrical-Electronics Engineering, Erzurum, 25240, Turkey

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ABSTRACT

Network slicing is a key mechanism for supporting heterogeneous vehicular services in 5G New Radio vehicle-to-everything (NR-V2X) systems. These services include latency-critical ultra-reliable low-latency communication (URLLC), throughput-oriented enhanced mobile broadband (eMBB), and event-driven vehicular safety messaging. In vehicular networks, however, traffic demand, channel quality, and slice resource needs can change rapidly because of mobility, fading, and bursty safety events. Therefore, slicing methods based mainly on stationary queueing assumptions or average-delay metrics may miss short congestion periods that affect URLLC tail latency, i.e., the delay experienced by the worst-served packets rather than the average packet delay.

This paper proposes a transient, deadline-aware radio access network (RAN) slicing framework for NR-V2X systems. The framework models URLLC, eMBB, and a logical Safety traffic class as finite-capacity time-varying queues, where the physical resource block (PRB) allocation of each slice directly determines its effective packet-level service parallelism. The proposed model combines mobility-driven arrivals, fading-dependent service variation, deadline-aware queue handling, predictive deadline dropping, and a tail-guided PRB controller. The controller adapts slice capacity using sliding-window URLLC p_{99} delay, defined as the delay value below which 99% of completed URLLC packets leave the serving next-generation NodeB (gNB) queue, together with queue congestion and short-horizon backlog pressure. Safety traffic is further evaluated through a separate direct sidelink interface (PC5) and hybrid automatic repeat request (HARQ) delivery stage, so that gNB queueing effects and sidelink delivery effects can be reported separately.

Simulation results show that the proposed framework stabilizes URLLC tail latency under representative NR-V2X conditions. In the representative scenario, URLLC p_{99} delay remains between 0.85 and 0.92 ms across the two gNBs, while the finite-horizon URLLC timely-completion ratio reaches 99.37% with 0.63% effective queue loss. The eMBB slice sustains approximately 12–15 Mb/s under nominal operation with no observed queue-level loss, and the PC5/HARQ delivery performance of the Safety traffic class is reported separately. Comparisons with static, adaptive, priority-based, and MaxWeight baselines show that transient queue modeling, predictive deadline dropping, and tail-guided PRB adaptation jointly improve URLLC tail-latency control while preserving broadband service. The results show that transient, deadline-aware RAN slicing can protect URLLC tail performance under time-varying NR-V2X conditions while maintaining broadband service.

1. Introduction

Connected and automated vehicles generate several types of wireless traffic at the same time. Periodic awareness messages report vehicle status, event-driven safety messages warn nearby users about hazards, cooperative driving messages support coordinated maneuvers, and infotainment traffic requires broadband data delivery. In 5G New Radio vehicle-to-everything (NR-V2X) systems, these heterogeneous services share the same radio access network (RAN), although they impose different latency, reliability, mobility, and throughput requirements [1–4].

Network slicing provides the 5G architectural mechanism for supporting these different services over common infrastructure and shared radio resources [5–7]. By assigning service-specific logical slices, the network can treat latency-critical ultra-reliable low-latency communication (URLLC), broadband enhanced mobile broadband (eMBB), and safety-oriented vehicular messages differently. The resulting RAN control problem is therefore not simply to allocate physical resource blocks (PRBs), but to preserve latency-critical performance while multiple traffic classes compete for finite radio and processing capacity.

* Corresponding author.
E-mail addresses: nurgunes.onal@atauni.edu.tr (N. Yordanov), bcavusoglu@atauni.edu.tr (B. Çavuşoğlu).

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This problem is especially difficult in vehicular networks because neither traffic demand nor effective service capability remains constant over time. Vehicle mobility changes radio conditions and network topology, while event-driven messages can create short congestion periods that are not well represented by long-term average-load indicators [8–10]. In this paper, *transient* refers to such time-varying operation, where arrivals, queue lengths, channel conditions, and service capacity change during the observation period instead of remaining fixed. At the queueing level, these variations can produce delay behavior that differs from steady-state predictions [11–14]. For latency-critical traffic, this behavior is decisive: the average delay may look acceptable even when a small fraction of packets waits too long and violates a strict deadline.

For this reason, this paper evaluates URLLC latency mainly through the 99th-percentile delay, denoted by URLLC p_{99}. In this paper, p_{99} is the time by which 99% of completed packets have left the serving next-generation NodeB (gNB) queue. This metric captures near-worst-case queueing behavior more clearly than the mean delay, because it shows whether almost all URLLC packets meet the required delay bound. The 1 ms target is used as a stringent gNB-side queueing and scheduling objective motivated by the low-latency nature of URLLC.

Latency must also be interpreted together with reliability. URLLC service requirements are commonly associated with both very low latency and high reliability, while V2X service requirements include stringent timing constraints for safety-relevant communication [2,3,15]. In the present scheduler-level framework, reliability is not modeled as a single physical-layer decoding probability. Instead, the paper evaluates queue-side reliability-relevant outcomes: whether an arriving packet can be admitted, served, and completed within its deadline.

This distinction is important because completed-packet delay percentiles describe only packets that have already departed from the queue. They do not show how many packets were blocked before admission, dropped because their deadline expired, removed because they could no longer finish in time, completed too late, or left unfinished at the end of the observation horizon. Accordingly, this paper reports three complementary metric groups. Completed-packet delay percentiles, represented in this paper by p_{99} (or more generally $p_{99,s}$ for slice s), quantify the tail delay of departed packets. Finite-horizon timely completion measures the fraction of all arriving packets that are completed within their deadline during the simulated observation period. Explicit loss decomposition identifies why the remaining packets fail, separating admission blocking, deadline expiration, predictive infeasibility drops, late service, and unfinished-by-horizon backlog. Together, these metrics provide a queue-side view of latency and reliability without replacing physical-layer or full end-to-end reliability analysis.

The model considered here is a scheduler-level abstraction aimed at slice-level control analysis. End-to-end URLLC performance also depends on lower-layer coding, link adaptation, retransmission design, physical-channel behavior, and complete radio/core-network procedures [16–20]. Therefore, the reported 1 ms result should be interpreted as a gNB-side queueing and scheduling URLLC p_{99} result, complemented by queue-side reliability-relevant metrics, including timely completion and loss decomposition.

1.1. Queueing model interpretation

This subsection clarifies how the queueing notation is interpreted in the proposed scheduler-level model. The notation follows the conventional Kendall-style description of arrival processes, service-time models, server counts, and finite capacity [21,22]. In a classical $M/M/c$ queue, the first M denotes a Poisson arrival process, the second M denotes exponential service times, and c denotes a fixed number of parallel servers. The model $M/M/c/K$ additionally includes a finite system capacity K, which counts both waiting and in-service packets.

The proposed slice queue model is denoted by

$$M(t)/G_{k_s}/c_s(t)/K_s(t).$$

Here, $M(t)$ denotes a time-varying Poisson arrival process, G_{k_s} denotes a general service-time distribution with slice-dependent shape parameter k_s, $c_s(t)$ denotes the time-varying number of effective packet servers assigned to slice s, and $K_s(t)$ denotes the finite capacity of that slice. Unlike a classical fixed-server queue, $c_s(t)$ is not constant. In the proposed abstraction, the PRB allocation of slice s determines how many packets can be served in parallel. Therefore, PRB reallocation changes the packet-level service capability of the slice rather than only an abstract average service rate.

Each admitted packet is also associated with a deadline. Earliest-deadline-first (EDF) scheduling means that the packet with the closest absolute deadline is served first. This rule keeps the service order aligned with packet urgency and is widely used in real-time queueing models [23,24].

1.2. Proposed approach and novelty

To address the above problem, this paper develops a transient, deadline-aware queueing and simulation framework for multi-slice NR-V2X RAN control. URLLC, eMBB, and a logical Safety traffic class are modeled as finite-capacity $M(t)/G_{k_s}/c_s(t)/K_s(t)$ queues with deterministic deadlines, deadline-aware service ordering, predictive deadline dropping, and slice-specific service-time variability. Because $c_s(t)$ is determined by the PRBs allocated to slice s, the framework directly connects radio-resource allocation to packet-level service parallelism. The Safety class represents event-driven vehicular messages and is treated as a simulator-level logical traffic class rather than as a formal 3GPP slice category. Safety traffic is additionally coupled with a PC5 sidelink/HARQ delivery stage, allowing gNB queueing effects and sidelink delivery effects to be reported separately.

The main novelty of the proposed framework lies in combining transient finite-capacity slice queues, deterministic deadline-aware packet handling, PRB-driven time-varying service parallelism, and tail-guided PRB adaptation within one causal simulation model. The controller uses sliding-window URLLC p_{99}, queue congestion, short-horizon backlog pressure, smoothing, and emergency reweighting to adapt the PRB split. This design keeps the analysis centered on slice-level mechanisms while still reflecting mobility- and channel-driven variation in effective service behavior.

1.3. Contributions

The main contributions of this paper are summarized as follows.

1. **Transient deadline-aware NR-V2X RAN slicing model:** A transient, deadline-aware RAN slicing model is formulated for NR-V2X systems. URLLC, eMBB, and a logical Safety traffic class are represented as finite-capacity $M(t)/G_{k_s}/c_s(t)/K_s(t)$ slice queues with non-stationary arrivals, deterministic deadlines, deadline-aware service ordering, predictive deadline dropping, and time-varying service capacity.
2. **PRB-driven packet-level service parallelism:** PRB allocation is modeled as packet-level service parallelism rather than only as an abstract service-rate or resource-share parameter. In the proposed abstraction, the PRB allocation of slice s directly determines the effective number of packet servers $c_s(t)$. Therefore, reallocating PRBs changes how many packets can be served in parallel within each slice.
3. **Predictive deadline dropping with tail-guided PRB control:** Predictive deadline dropping is combined with a tail-guided PRB controller. Packets that are unlikely to finish before their deadlines are removed before consuming service capacity, while slice capacity is adapted according to URLLC p_{99} delay risk, queue congestion, short-horizon backlog pressure, smoothing, and emergency reweighting. This combination is used to react to emerging URLLC tail-latency violations before they become persistent queueing failures.

4. **Finite-horizon timely-completion and loss decomposition metrics:** Performance is evaluated not only through completed-packet delay percentiles, but also through finite-horizon timely-completion and explicit loss-decomposition metrics. In particular, queue-side indicators such as R_{e2e}, P_{eff}, P_{dead}, P_{late}, and P_{pend} are reported to distinguish timely delivery, effective queue loss, deadline-induced loss, late service, and unfinished-by-horizon backlog.
5. **Separate PC5/HARQ evaluation for Safety traffic:** The logical Safety traffic class is evaluated with a separate downstream PC5 sidelink/HARQ delivery abstraction. This allows gNB-side queueing behavior to be reported separately from sidelink delivery performance, so that queue-side losses and PC5/HARQ delivery effects are not mixed into a single reliability value.
6. **Baseline comparison and performance evidence:** The proposed framework is compared with static, adaptive, priority-based, and MaxWeight-type slicing baselines. In the representative NR-V2X scenario, URLLC p_{99} delay is kept between 0.85 and 0.92 ms across the two gNBs, while a 99.37% finite-horizon URLLC timely-completion ratio is achieved with 0.63% effective URLLC queue loss. The results show that transient queue modeling, predictive deadline dropping, and tail-guided PRB adaptation jointly stabilize URLLC tail latency while preserving broadband service.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and positions the present study. Section 3 introduces the system model and the overall methodological workflow. Section 4 presents the slice queueing architecture, deadline-handling mechanisms, and performance metrics. Section 5 describes the gNB model and the tail-guided dynamic PRB controller. Section 6 defines the channel, mobility, and PC5 sidelink abstractions. Section 7 reports the main simulation results, while Section 8 provides the comparison with representative slicing baselines. Section 9 discusses the interpretation, modeling scope, and deployment-oriented implications of the proposed framework. Finally, Sections 10 and 11 conclude the paper and outline directions for future work.

2. Related works

The problem studied in this paper lies at the intersection of vehicular network slicing, URLLC–eMBB resource sharing, deadline-aware queueing, and transient service dynamics. Prior studies have advanced each of these areas substantially, but they usually focus on different layers of the problem, such as slicing architecture, long-term resource allocation, learning-based adaptation, or analytically tractable queueing models. As a result, transient finite-capacity queue evolution, predictive deadline-aware packet handling, and PRB-driven changes in packet-level service parallelism are rarely examined within one scheduler-level NR-V2X framework. This section reviews the literature from that perspective and identifies the gap addressed by the proposed transient, deadline-aware, PRB-controlled slicing model.

2.1. NR-V2X service requirements and vehicular network slicing

NR-V2X systems are designed to support a wide range of vehicular services, including cooperative awareness, event-driven safety notification, automated driving, platooning, and high-throughput infotainment. The service diversity is reflected in 3GPP and ETSI specifications, where V2X requirements include both safety and non-safety use cases with different latency, reliability, mobility, and communication-pattern constraints [1–3,15]. ETSI CAM and DENM specifications further illustrate the coexistence of periodic awareness traffic and event-triggered warning traffic, which motivates a system model that distinguishes regular vehicular updates from bursty safety notifications [25,26]. At the system-architecture level, 5G slicing provides the mechanism for mapping heterogeneous services onto logical network instances over a shared infrastructure [5].

Survey and architectural studies on NR-V2X and vehicular slicing provide the foundation for this view. Bagheri et al. review the evolution of NR-V2X toward cooperative and connected autonomous driving, emphasizing sidelink operation, flexible architecture, positioning, and advanced vehicular use cases [4]. Campolo et al. introduce the concept of 5G network slicing for V2X services and show how service-specific logical networks can be tailored to vehicular requirements [6]. More recent works extend this direction by considering learning-assisted slicing for the Internet of Vehicles (IoV), QoS-aware resource management, practical slice allocation frameworks for 5G V2X deployments, 5G/RSU-integrated V2X resource allocation, AI-driven O-RAN slicing for connected vehicles, and transient edge-slicing support for connected autonomous vehicles [27–32].

These studies are essential for defining the architectural and service-management context of vehicular slicing. Their primary focus, however, is not packet-level transient queue control. In particular, they generally do not model how short congestion episodes propagate through finite queues, how deterministic packet deadlines create deadline-induced loss, or how a change in the PRB budget modifies the number of packets that can be served concurrently. The present work builds on the slicing perspective of these studies but shifts the modeling focus from slice creation and service mapping to transient deadline-aware operation inside the RAN scheduler.

2.2. URLLC–eMBB multiplexing and PRB-level RAN resource sharing

A second line of work studies the coexistence of URLLC and eMBB over shared radio resources. Popovski et al. analyze 5G slicing for eMBB, URLLC, and mMTC from a communication-theoretic perspective and show that heterogeneous reliability requirements can be exploited through orthogonal and non-orthogonal slicing strategies [7]. Pocovi et al. examine joint link adaptation and scheduling for URLLC and eMBB multiplexing, highlighting the need to coordinate reliability, scheduling, and radio adaptation under stringent latency constraints [17]. Korrai et al. formulate RAN slicing for multiplexing eMBB and URLLC services in OFDMA-based 5G systems, while Bakri et al. propose dynamic RAN resource slicing for heterogeneous coexisting services [33,34]. Related physical-layer slicing studies also consider rate-splitting multiple access for URLLC uplink coexistence with eMBB [35]. Broader surveys and recent scheduling formulations summarize QoS-aware coexistence mechanisms, weighted scheduling strategies, and service-assurance-oriented pre-emptive slicing mechanisms for next-generation cellular networks [36–38].

These works establish that URLLC and eMBB coexistence is fundamentally a resource-sharing problem. Nevertheless, many such models evaluate resource allocation through average delay, throughput, outage, long-term QoS satisfaction, or equilibrium resource shares. These metrics are informative but incomplete for deadline-critical URLLC operation because a low average delay can coexist with rare but severe tail-latency excursions. Moreover, PRB allocation is often represented as an abstract rate or resource fraction. In contrast, the present work explicitly maps the PRB allocation of slice s to a packet-server count $c_s(t)$, so that radio reallocation directly changes the number of packets that can be served in parallel. This allows the scheduler to evaluate not only whether a slice receives more resources, but also whether those resources are sufficient to prevent imminent deadline violations.

2.3. Learning-based and optimization-based RAN control

Learning-based control has become an important tool for adaptive slicing and RAN resource management. Li and Zhang use deep reinforcement learning for joint scheduling of eMBB and URLLC traffic, targeting the QoS trade-off between heterogeneous services [39]. Filali et al. propose a dynamic SDN-based RAN slicing framework with deep reinforcement learning for URLLC and eMBB services, using different control time scales for resource allocation [40]. Firouzi and

Rahmani address delay-sensitive resource allocation in 5G O-RAN systems through hierarchical learning-based control, while Abba Ari et al. survey learning algorithms for resource allocation and slicing orchestration in IoT-5G and B5G/6G systems [41,42]. Recent O-RAN-oriented studies further investigate QoS-aware slice resource allocation with user association parameterization, hierarchical learning for Open RAN slicing, and AI-driven slicing for connected-vehicle connectivity [31,43,44]. These works demonstrate that learning-based controllers can adapt to time-varying traffic and network conditions without requiring a closed-form analytical model.

However, learning-based slicing usually depends on reward design, state representation, and scenario-specific training. In many cases, the optimization signal is based on average throughput, average delay, queue length, or aggregate QoS satisfaction. Such objectives may not explicitly encode deterministic packet deadlines, earliest-deadline-first service, predictive deadline infeasibility, or loss decomposition. For URLLC-oriented vehicular operation, this distinction is important: a controller can improve average reward while still allowing packets close to their deadlines to remain in the queue until they become unrecoverable. The framework proposed in this paper is therefore complementary to learning-based RAN control. It provides an interpretable transient model in which the control signal is directly tied to sliding-window URLLC p_{99}, finite queue occupancy, deadline expiration, and PRB-driven service feasibility.

2.4. Queueing, deadline-aware scheduling, and URLLC resource allocation

Queueing theory provides the analytical basis for modeling congestion, service capacity, loss, and stability in communication systems. Classical queueing references and matrix-analytic methods define the notation and modeling tools for finite-capacity queues, service processes, and loss systems [21,22]. In wireless scheduling, the stability results of Tassiulas and Ephremides established the importance of queue-aware scheduling policies for constrained radio networks [45]. Heavy-traffic and many-server approximations further show how service capacity, abandonment, and congestion interact in large-scale service systems [46,47].

For URLLC and low-latency wireless access, queueing models have been used to study resource allocation, HARQ design, proactive dropping, reliability constraints, and coexistence between short-deadline and broadband traffic. Anand and de Veciana analyze OFDMA resource allocation and HARQ optimization for URLLC traffic using a queueing-based model [18]. Jang et al. study URLLC-mode resource allocation with HARQ support [19]. Qureshi et al. investigate resource allocation for multi-device URLLC networks, while She et al. formulate a cross-layer URLLC optimization problem that explicitly accounts for transmission delay, queueing delay, and proactive packet dropping [48,49]. Guo et al. address vehicular resource allocation with latency and reliability constraints, and Esswie and Pedersen study preemptive scheduling for critical low-latency communications [50,51]. Recent service-assurance-oriented slicing work also shows that pre-emptive scheduling remains relevant for maintaining QoS differentiation in next-generation cellular networks [38].

Recent studies have also revisited slicing and delay-sensitive resource allocation from an explicitly queue-aware perspective. Chai and Wang formulate delay-guaranteed RAN slicing through physical and virtual queue backlogs within a Lyapunov optimization framework, jointly adapting RB and power allocation under time-varying traffic and channel conditions [52]. Yang et al. study deterministic-delay network slicing in MEC by modeling the system as a many-to-one cascade queue composed of UE offloading queues and an edge computation queue [53]. Jin et al. show that average-delay optimization and deterministic delay constraints may require different queue-aware control structures, and they use virtual queues within a renewal-based resource-allocation framework [54]. In the V2X context, Liu et al. use deterministic network calculus to bound delay under non-orthogonal slicing for URLLC and eMBB services [55].

Deadline-aware queueing is also closely related to real-time queue theory. Doytchinov et al. analyze real-time queues under earliest-deadline-first service in heavy traffic, and Atar et al. develop a many-server earliest-deadline-first queueing limit with general service-time and deadline distributions [23,24]. These studies show that deadline ordering is not merely an implementation detail; it changes the structure of the queueing process and the interpretation of delay performance.

Compared with the above studies, the present work focuses on a different operating regime and modeling level. Instead of deriving closed-form delay or stability expressions under stationary traffic or fixed service capacity, it uses a finite-capacity event-level simulation in which packets may be admitted, blocked, served, expired, or dropped by a predictive deadline-feasibility test. More importantly, PRB allocation is represented as a change in packet-level service parallelism through $c_s(t)$, rather than only as an abstract rate, utility, or resource-share variable. This distinction is important because URLLC performance is not determined only by the latency of completed packets. Packets that are blocked at admission, expire while waiting, or become infeasible before service must also be counted as reliability-relevant losses.

2.5. Time-varying queueing models and transient congestion

Classical steady-state queueing models are limited when the arrival process and service capacity change over time. Eick, Massey, and Whitt show that the behavior of $M_t/G/\infty$ queues depends strongly on the arrival-rate function and the service-time distribution, and their sinusoidal-arrival analysis demonstrates that transient queue behavior may lag behind or differ substantially from the instantaneous offered load [11,12]. Massey further discusses queues with time-varying rates in telecommunication models [13]. Liu and Whitt extend the time-varying view to networks of many-server fluid queues with customer abandonment and stabilization under time-varying arrivals [56,57]. Later work on time-varying and robust queueing emphasizes that transient workload and high-percentile delay behavior cannot always be inferred from stationary approximations [14,58]. Non-stationary loss-queue approximations similarly show the need to evaluate finite-capacity systems under time-dependent demand [59]. Recent edge-slicing work for connected autonomous vehicles further confirms that transient resource provisioning is an active problem in vehicular networked systems [32].

These results are directly relevant to NR-V2X slicing because vehicular traffic and radio service capacity are rarely stationary. Mobility, fading, handover-like association changes, safety events, and bursty application demand can produce short-lived overload intervals. Nevertheless, standard time-varying queueing models do not directly include the combination required here: finite slice-specific capacities, deterministic packet deadlines, EDF scheduling, PRB-induced changes in packet-level parallelism, fading-aware service variation, and a feedback controller driven by URLLC tail delay. The present paper therefore uses time-varying queueing theory as the conceptual basis for the model, while relying on discrete-event simulation to capture the interactions that become analytically intractable under realistic vehicular scheduling assumptions.

2.6. Mobility, sidelink, and simulation-oriented vehicular modeling

A realistic vehicular slicing model also requires mobility and radio abstractions. Mobility-model studies show that node movement assumptions can strongly affect wireless simulation outcomes [8,9]. High-fidelity vehicular communication simulation has also been shown to be important when evaluating intersection-level control and V2X behavior [10]. At the radio level, NR physical-channel specifications define the underlying structure of channels and modulation, while recent link-level NR-V2X sidelink studies quantify how sidelink communication behaves under practical physical-layer assumptions [16,

20]. Recent connected-vehicle and 5G/RSU slicing studies also emphasize that vehicular resource allocation must account for heterogeneous QoS requirements across infrastructure-assisted and roadside-assisted communication settings [30,31].

The proposed framework does not attempt to replace full link-level or network-level NR-V2X simulation. Instead, it adopts a scheduler-layer abstraction designed to expose the control mechanism under study. The gNB-side queues model slice-level contention over PRBs, while the safety-oriented logical traffic class includes a PC5/HARQ delivery abstraction to separate RAN queueing effects from sidelink delivery effects. This separation is useful because a safety packet may avoid gNB queue loss but still fail at the sidelink delivery stage. Reporting these effects separately prevents the reliability metric from hiding the source of loss.

2.7. Comparison with existing work

The reviewed literature shows that the main components of the problem have largely been studied in isolation: vehicular slicing defines the architectural need for service differentiation, URLLC–eMBB coexistence studies analyze radio-resource sharing, deadline-aware queueing formalizes real-time service ordering, and time-varying queueing theory explains why transient behavior can differ from steady-state predictions. The main gap that remains at the scheduler level is the lack of a unified framework that jointly captures transient finite-queue dynamics, deterministic deadline enforcement, PRB-induced packet-level service parallelism, and tail-latency-driven slice adaptation. Accordingly, the objective of this paper is to develop a transient deadline-aware NR-V2X RAN slicing framework that addresses this gap through finite-capacity queue modeling, predictive deadline handling, and tail-guided PRB adaptation.

Table 1 provides a compact comparison between representative research streams and the proposed framework.

3. System model and methodological workflow

This section introduces the overall system model before presenting the detailed queueing equations. The goal is to make explicit the problem setting, the modeled entities, the main assumptions, and the causal workflow of the proposed simulator and controller.

3.1. Network setting and slice roles

We consider a multi-gNB NR-V2X radio access network in which vehicle-mounted user equipments (UEs) generate heterogeneous traffic and share the radio resources of their serving gNBs. Each gNB manages a fixed physical resource block (PRB) budget and partitions this budget among three logical slices: URLLC, eMBB, and Safety. The same slice-level control logic is executed independently at each gNB using local queue, delay, and PRB-allocation states.

The URLLC slice represents latency-critical traffic with a tight queueing and scheduling deadline. Its main control objective is to maintain the slice-level 99th-percentile delay, p_{99}, below the 1 ms target.

The eMBB slice represents broadband vehicular traffic and is evaluated mainly through sustained throughput and queue-level loss.

The Safety slice represents event-driven vehicular safety messages. It is modeled as a separate logical slice because its delivery process includes both gNB-side queueing and a downstream PC5 sidelink/HARQ delivery stage. Although 3GPP formally defines URLLC, eMBB, and mMTC service categories, NR-V2X studies commonly group vehicular traffic into safety, traffic-efficiency, and infotainment classes [1,25,26]. Following this convention, the present framework includes a logical “Safety” slice for medium-sized event-driven NR-V2X messages such as CAM/DENM, without treating it as a formal 3GPP slice type.

The three slices differ in their packet deadlines, service-time distributions, PRB requirements, and performance objectives, but they share the same PRB pool at each gNB. The slicing problem is therefore to adapt the PRB allocation $n_s(t)$, and hence the effective number of packet servers $c_s(t)$, so that URLLC tail latency is controlled while preserving eMBB service and Safety delivery performance.

3.2. System workflow and queue-control interaction

Fig. 1 summarizes the methodological workflow. The simulator evolves as a causal discrete-event system. First, vehicle mobility and traffic-generation processes produce time-varying slice arrivals. Second, the radio/channel abstraction maps mobility and fading conditions into effective PRB and service conditions. Third, packets are classified into URLLC, eMBB, and Safety slices at the serving gNB. Fourth, each slice operates as a finite-capacity, deadline-aware $M(t)/G_{k_s}/c_s(t)/K_s(t)$ queue. Packets are admitted subject to finite-capacity availability; for URLLC, a lightweight controller-side admission guard is additionally applied. Waiting packets are served according to earliest-deadline-first (EDF), and deadline feasibility is checked again when service is about to begin, so packets that can no longer finish on time are dropped before consuming server time.

At each control instant, the gNB updates recent URLLC sampled mean-sojourn and queue-occupancy signals, computes the controller-side URLLC tail estimate $\hat{p}_{99,\mathrm{ctrl}}(t)$, and evaluates URLLC unfinished-arrival pressure together with Safety queue/tail stress before updating the slice PRB allocation. The new PRB allocation changes $c_s(t)$, the number of effective packet servers available to slice s, and therefore changes the future service capacity of that slice. This feedback loop captures the key transient interaction between vehicular load, queueing delay, deadline feasibility, and radio-resource allocation.

3.3. Main modeling assumptions

Table 2 summarizes the main modeling assumptions used in the framework. These assumptions define the intended scope of the model. The simulator is designed to evaluate slice-level queueing, scheduling, deadline feasibility, and PRB-control behavior; it is not intended to replace a symbol-level physical-layer or full protocol-stack simulator.

3.4. Notation and compact parameter summary

Before presenting the detailed queueing and control equations, Tables 3 and 4 summarize the main notation and parameter settings used throughout the manuscript. The notation distinguishes between queue-side finite-horizon timely-completion, effective queue loss, deadline-induced queue loss, late service, unfinished-by-horizon backlog, and downstream PC5/HARQ delivery reliability, which is important for interpreting the latency–reliability–loss–backlog trade-off reported in the results.

4. Slice queueing architecture

4.1. Time-varying deadline-aware queues

We adopt a transient queueing perspective because both offered load and effective service capacity vary over time in vehicular NR-V2X systems [11,47]. Classical $M/M/c$ models provide an important baseline, but their steady-state assumptions are not well aligned with mobility-driven traffic variation, bursty safety events, and adaptive PRB reallocation.

Eick, Massey, and Whitt studied time-varying queues of the form

$$M(t)/M(t)/1, \qquad M(t)/M(t)/c(t), \tag{1}$$

and showed that queue evolution depends strongly on the temporal shape of arrivals and service capacity [11,14]. Later work extended this perspective to $M(t)/G(t)/c(t)$ systems and showed that transient high-percentile delay behavior is driven by short-term overload episodes, while features such as finite buffers, general service times, and abandonment are typically handled through numerical or simulation-based analysis [46,56,57].

Table 1
Representative research streams and how the proposed framework differs.

Research stream and representative examples	Main limitation for the problem considered here	How the proposed framework differs
NR-V2X requirements, service mapping, and vehicular slicing [1–6,28–31]	These works define V2X service classes, slice architectures, orchestration frameworks, and QoS-aware vehicular slicing mechanisms. However, they are mainly architectural or system-management oriented and usually do not study scheduler-level transient finite-queue evolution, deterministic deadline loss, or packet-level admission/service dynamics.	The proposed framework moves the focus inside the RAN scheduler. It models URLLC, eMBB, and a logical Safety class as finite-capacity queues with explicit deadlines, queue-side timely-completion accounting, and loss decomposition under time-varying vehicular load.
URLLC–eMBB multiplexing and PRB/RB-level RAN resource sharing [7,17,33–38]	These studies analyze coexistence, puncturing, link adaptation, weighted scheduling, and service-assurance trade-offs, but evaluation is often based on average delay, throughput, outage, or long-term QoS. PRB allocation is also frequently represented as an abstract rate or resource share.	The proposed framework uses sliding-window URLLC tail delay, deadline feasibility, finite-horizon timely completion, and explicit queue-loss decomposition as scheduler-level signals. It further maps PRB allocation directly to packet-level service parallelism through $c_s(t)$.
Learning-based and O-RAN-oriented adaptive control [31,39–44]	Learning-based methods adapt well to changing traffic and radio conditions, but their behavior depends on reward design, state representation, and scenario-specific training. Deterministic packet deadlines, infeasible-packet removal, and interpretable deadline-loss semantics are often implicit rather than explicit.	The proposed controller is interpretable and causal. It directly uses measured queue occupancy, sampled delay, controller-side URLLC tail estimates, and unfinished-work pressure, while preserving explicit deadline-aware packet handling and transparent queue-side metrics.
Queueing-based URLLC, deterministic-delay, and low-latency scheduling [18,19,48–55]	These works study delay bounds, HARQ-aware design, proactive dropping, deterministic-delay allocation, virtual queues, and pre-emptive scheduling, but many rely on stationary traffic, fixed capacity, analytical tractability, or more abstract service models than the present scheduler-level setting.	The proposed model evaluates event-level transient behavior with finite buffers, general service-time variability, deterministic deadlines, predictive feasibility drops, and adaptive PRB allocation under vehicular load variation.
EDF and real-time queueing [23,24]	EDF and real-time queueing studies provide the theoretical structure of deadline-ordered service, but they do not directly address NR-V2X slicing, PRB-controlled service parallelism, mobility-driven radio variation, or sidelink delivery stages.	The proposed framework embeds EDF-like deadline-aware service in a vehicular RAN slicing simulator and combines it with finite-capacity admission, PRB control, and explicit queue-side reliability-relevant metrics.
Time-varying queueing and transient resource provisioning [11–14,32,56–59]	These works explain why non-stationary arrivals and service capacity can produce transient delay behavior, but they do not directly provide a vehicular slicing model with deterministic packet deadlines, slice-specific PRB control, and queue-side deadline-feasibility enforcement.	The proposed framework adopts a transient $M(t)/G_{k_s}/c_s(t)/K_s(t)$-style slice abstraction in which $c_s(t)$ is controlled by PRB allocation and queue-side tail-delay risk is used in the control loop.
Mobility, sidelink, and V2X simulation realism [8–10,16,20,30,31]	These studies improve realism at the mobility, radio, or sidelink level, but they often focus on physical/link simulation or communication realism rather than tail-latency-driven slice control with explicit queue-side loss accounting.	The proposed framework keeps mobility and fading effects through a scheduler-layer abstraction while isolating the RAN slicing mechanism and reporting gNB queueing outcomes separately from Safety PC5/HARQ delivery outcomes.

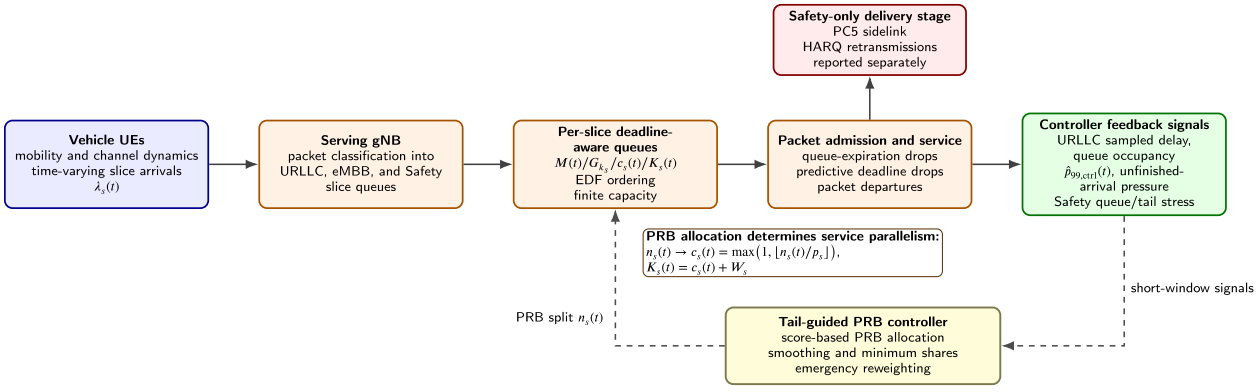


Fig. 1. The controller uses short-horizon queue-side signals, including sampled URLLC mean sojourn, URLLC queue occupancy, the controller-side URLLC tail estimate $\hat{p}_{99,\mathrm{ctrl}}(t)$, URLLC unfinished-arrival pressure, and safety queue/tail stress, and reallocates PRBs to modify the effective number of packet servers $c_s(t)$. Safety traffic additionally includes a PC5 sidelink/HARQ stage whose delivery metrics are reported separately from the main queue-side slice metrics.

Even $M(t)/G(t)/c(t)$ queues with infinite buffers rarely admit closed-form transient delay characterizations for high-percentile latency metrics [14,22]. In the present setting, finite capacity, deterministic deadlines, predictive deadline dropping, and PRB-driven changes in service parallelism make a causal discrete-event formulation the natural choice for evaluating tail delay and deadline-related queue behavior.

4.2. Service times and variance shaping

Each slice is modeled as a finite-capacity, multi-server, non-stationary, deadline-aware queue of class

$$M(t)/G_{k_s}/c_s(t)/K_s(t), \tag{2}$$

Table 2
Main assumptions of the proposed system model.

Component	Assumption and role in the model
Network entities	Vehicle-mounted user equipments (UEs) generate traffic and are served by next-generation NodeBs (gNBs). Each gNB executes the slice controller using local queue states, deadline states, and physical resource block (PRB) allocation states.
Logical traffic classes	Three logical traffic classes are considered: URLLC for latency-critical traffic, eMBB for broadband traffic, and Safety for event-driven vehicular safety messages. The Safety class is used as a simulator-level logical traffic class for modeling purposes.
Radio resources	Each gNB has a fixed PRB budget. The controller reallocates PRBs among traffic classes over time. In the queueing abstraction, PRB allocation changes the number of effective packet servers assigned to each class, rather than the service rate of an individual PRB.
Queueing model	Each traffic class is represented as a finite-capacity $M(t)/G_{k_s}/c_s(t)/K_s(t)$ queue with non-stationary arrivals, slice-specific service-time variability, deterministic deadlines, and EDF-based deadline-aware scheduling.
Deadline handling	Packets can be rejected by finite-capacity admission blocking, removed after deadline expiration in the queue, or dropped by predictive feasibility checking before service if they can no longer complete before their deadline.
Safety delivery	Safety traffic includes gNB-side queueing and a separate PC5 sidelink/HARQ delivery stage. Queue-side timely-completion and loss metrics are therefore reported separately from downstream PC5/HARQ delivery metrics.
Controller input	The PRB controller observes recent queue occupancy, mean sojourn delay, the controller-side URLLC tail-delay estimate $\hat{p}_{99,\mathrm{ctrl}}(t)$, unfinished-arrival pressure, and Safety queue/tail stress indicators. The corresponding controller and reporting windows are specified separately in the parameter tables.
Controller output	The controller computes slice PRB allocations $n_s(t)$. These allocations determine the effective packet-level service parallelism $c_s(t) = \max\left(1, \lfloor n_s(t)/p_s \rfloor\right)$, and hence the time-varying service capacity of each queue.
Performance metrics	The reported delay metric is the slice-specific $p_{99,s}$, with URLLC p_{99}, i.e., $p_{99,U}$, used as the primary tail-latency target. Additional reported metrics include the finite-horizon timely-completion ratio $R_{\mathrm{e2e},s}$, effective queue loss $P_{\mathrm{eff},s}$, capacity-induced loss $P_{\mathrm{cap},s}$, deadline-induced loss $P_{\mathrm{dead},s}$, late service $P_{\mathrm{late},s}$, pending backlog $P_{\mathrm{pend},s}$, eMBB throughput, and separate PC5/HARQ delivery reliability metrics for Safety traffic.
Modeling boundary	The framework is a scheduler-level abstraction for studying slice-level queueing, deadline handling, and PRB-control dynamics in NR-V2X systems. It is not intended as a symbol-level PHY or full protocol-stack simulator. Within this scope, lower-layer communication effects are represented through service-time distributions, PRB requirements, fading-dependent service variation, and the separate Safety PC5/HARQ delivery abstraction.

where $M(t)$ denotes a non-stationary Poisson arrival process with rate $\lambda_s(t)$, $c_s(t)$ denotes the number of effective packet servers available to slice s at time t, and

$$K_s(t) = c_s(t) + W_s \tag{3}$$

is the total slice capacity, with W_s denoting a fixed slice-specific waiting capacity.

A new arrival is admitted if

$$Q_s(t) < K_s(t), \tag{4}$$

where $Q_s(t)$ is the instantaneous number of packets in service and waiting. Otherwise, the packet is blocked.

Service times represent effective packet-completion times at the slice level, absorbing PHY/MAC transmission, scheduling delays, and retransmission effects into a single service requirement. To shape delay variability, especially for URLLC traffic, service times are modeled using Gamma/Erlang distributions:

$$S_s \sim \Gamma(k_s, \theta_s), \qquad \mathbb{E}[S_s] = \frac{1}{\mu_{0,s}}. \tag{5}$$

The shape parameter k_s controls service-time variance. For URLLC, $k_U > 1$ is used to reduce variability and improve tail behavior, whereas eMBB and Safety traffic use exponential service with $k_E = k_S = 1$. This variance shaping is an important degree of freedom in the transient URLLC queue design.

4.3. Deadlines and EDF scheduling

Each admitted packet is assigned an absolute deadline

$$\Theta_n = A_n + D_s, \tag{6}$$

where A_n is the packet arrival time and D_s is the slice-specific sojourn-time deadline. URLLC uses the tightest deadline, while eMBB and Safety use larger slice deadlines.

Packets waiting within a slice are ordered according to Earliest-Deadline-First (EDF) [23,24]:

$$\text{Serve the packet with } \min_j \Theta_j. \tag{7}$$

EDF is used here as a standard deadline-aware service rule that keeps service ordering aligned with packet urgency.

4.4. Deadline enforcement mechanisms

Two deadline-enforcement mechanisms are included.

Queue-expiration drops. A packet is removed if, at time t,

$$t \geq \Theta_j, \tag{8}$$

its deadline has already expired while it is still waiting.

Predictive deadline drops. When a packet is about to begin service at time t, a service-time sample S is drawn from the slice-specific service-time distribution. Service proceeds only if

$$t + S \leq \Theta_j. \tag{9}$$

Otherwise, the packet is dropped before consuming server time.

4.5. Implementation-level queue procedure

To make the event-level queue behavior explicit, Algorithm 1 summarizes the implemented per-slice queue procedure.

4.6. Performance metrics and SLA indicators

When packet n of slice s departs at time t, its slice-queue sojourn time is

$$W_{n,s} = t - A_{n,s}, \tag{10}$$

where $A_{n,s}$ is the packet arrival time. A departed packet is counted as timely if $W_{n,s} \leq D_s$.

For each slice $s \in \{U, E, S\}$, the simulator maintains the counters total arrivals N_s^{arr}, total departures N_s^{dep}, timely departures N_s^{ok}, aggregate blocked arrivals B_s, capacity-induced blocks $B_{\mathrm{cap},s}$, and aggregate deadline drops L_s^{dead}. Capacity-induced blocks occur when the finite slice capacity is already full at arrival. Deadline drops include both waiting-queue expiration and predictive deadline drops before service. Thus,

$$B_s = B_{\mathrm{cap},s} + L_s^{\mathrm{dead}}. \tag{11}$$

For interpretation, the aggregate deadline-induced queue-loss count is decomposed into queue-expiration losses $L_s^{(q)}$ and predictive deadline losses $L_s^{(p)}$:

$$L_s^{\mathrm{dead}} = L_s^{(q)} + L_s^{(p)}. \tag{12}$$

Table 3
Core queueing, deadline, timely-completion, tail-delay, and loss-decomposition notation.

Symbol	Meaning
t	Simulation time.
$T_{\max}$	Total simulation duration.
$\mathcal{S} = \{U, E, S\}$	Set of slices: URLLC (U), eMBB (E), and Safety (S).
s	Generic slice index, $s \in \mathcal{S}$.
$\lambda_s(t)$	Time-varying arrival rate of slice s.
$\lambda_s^{(0)}, a_s, \omega_s, \lambda_s^{(\text{burst})}(t)$	Nominal arrival rate, oscillation amplitude, angular variation parameter, and burst-arrival component of slice s.
$M(t)/G_{k_s}/c_s(t)/K_s(t)$	Finite-capacity, time-varying slice queue with non-stationary Poisson arrivals, general service times, PRB-induced packet-server count, and time-varying capacity.
k_s	Gamma/Erlang service-time shape parameter of slice s.
S_s, S_n	Slice-specific service-time random variable and the sampled service time of packet n.
$\mu_{0,s}$	Service rate of one effective packet server in slice s (packets/s).
$c_s(t) = \max\big(1, \lfloor n_s(t)/p_s \rfloor\big)$	PRB-induced number of effective packet servers in slice s.
W_s	Fixed waiting capacity of slice s.
$K_s(t)$	Total slice capacity, defined as $K_s(t) = c_s(t) + W_s$.
A_n	Arrival time of packet n.
D_s	Slice-specific sojourn-time deadline.
Θ_n	Absolute deadline of packet n, $\Theta_n = A_n + D_s$.
$Q_s(t)$	Instantaneous number of packets waiting or in service in slice s.
W_n	Sojourn time of packet n, measured from arrival to queue departure.
$\overline{Q}_s(t)$	Sliding-window mean queue occupancy of slice s.
$\overline{W}_s(t)$	Sliding-window mean sojourn time of departed packets in slice s.
T_w	Generic sliding-window length used for empirical queueing and delay measurements.
T_w^{ctrl}	Controller-side tail-delay estimation window. In the simulations, $T_w^{\text{ctrl}} = 5$ s.
T_w^{rep}	Reporting-window length used for plotted URLLC tail-delay traces. In the simulations, $T_w^{\text{rep}} = 10$ s.
p_{99}	Reported URLLC 99th-percentile gNB-side queueing sojourn delay over the selected evaluation set or simulation horizon.
$p_{99,s}$	Reported slice-specific gNB-side queueing 99th-percentile sojourn delay for slice $s \in \mathcal{S}$. In this paper, $p_{99,U}$ is written simply as p_{99}, whereas $p_{99,E}$ and $p_{99,S}$ are used only when eMBB or Safety queueing percentiles are explicitly reported.
$\hat{p}_{99}(t; T_w)$	Empirical sliding-window estimate of the URLLC 99th-percentile gNB-side queueing sojourn delay at time t, computed over departed URLLC packets in the window $(t - T_w, t]$.
$\hat{p}_{99,\text{ctrl}}(t)$	Controller-side URLLC tail-delay estimate, defined as $\hat{p}_{99,\text{ctrl}}(t) = \hat{p}_{99}(t; T_w^{\text{ctrl}})$. This estimate is used by the PRB controller, emergency logic, and admission-related safeguards.
$\hat{p}_{99,\text{rep}}(t)$	Reporting-window URLLC tail-delay estimate, defined as $\hat{p}_{99,\text{rep}}(t) = \hat{p}_{99}(t; T_w^{\text{rep}})$. This estimate is used for plotted URLLC p_{99} traces.
$\dot{p}_{99}(t)$	Short-term slope of the controller-side URLLC tail-delay estimate $\hat{p}_{99,\text{ctrl}}(t)$, estimated from recent controller samples and used for emergency detection. The unit is ms/s.
$\dot{p}_{\text{emg}}$	Emergency threshold applied to $\dot{p}_{99}(t)$.
N_s^{arr}	Total number of arrivals to slice s over the observation horizon.
N_s^{dep}	Total number of departed packets from slice s over the observation horizon.
N_s^{ok}	Number of timely departures from slice s, i.e., departed packets satisfying the slice deadline.
$B_{\text{cap},s}$	Number of structural admission blocks caused by finite slice capacity.
$L_s^{(q)}$	Number of queue-expiration losses, i.e., packets whose deadline expires while waiting.
$L_s^{(p)}$	Number of predictive-feasibility losses, i.e., packets dropped before service because completion before the deadline is no longer feasible.
B_s	Aggregate queue-loss counter of slice s, defined as $B_s = B_{\text{cap},s} + L_s^{(q)} + L_s^{(p)}$.
L_s^{dead}	Aggregate deadline-induced queue-loss counter of slice s, defined as $L_s^{\text{dead}} = L_s^{(q)} + L_s^{(p)}$.
$P_{\text{eff},s}$	Effective queue loss of slice s, defined as the aggregate queue-loss fraction $P_{\text{eff},s} = B_s/N_s^{\text{arr}}$.
$P_{\text{cap},s}$	Capacity-induced queue-loss fraction of slice s, defined as $P_{\text{cap},s} = B_{\text{cap},s}/N_s^{\text{arr}}$.
$P_{\text{dead},s}$	Deadline-induced queue-loss fraction of slice s, defined as $P_{\text{dead},s} = L_s^{\text{dead}}/N_s^{\text{arr}}$.
$P_{\text{late},s}$	Late-served fraction of slice s, defined as $P_{\text{late},s} = (N_s^{\text{dep}} - N_s^{\text{ok}})/N_s^{\text{arr}}$.
$P_{\text{pend},s}$	Unfinished-by-horizon backlog fraction of slice s, defined as $P_{\text{pend},s} = (N_s^{\text{arr}} - B_s - N_s^{\text{dep}})/N_s^{\text{arr}}$.
$R_{\text{cond},s}$	Completed-packet conditional timely-completion ratio of slice s, computed over departed packets only.
$R_{\text{e2e},s}$	Queue-side finite-horizon timely-completion ratio of slice s, computed over all arrivals in the observation horizon.
Tail_s	Finite-horizon non-timely-completion fraction of slice s, defined as $\text{Tail}_s = 1 - R_{\text{e2e},s}$.

Here, $L_s^{(q)}$ counts packets whose deadlines expire while waiting, whereas $L_s^{(p)}$ counts packets dropped immediately before service when the sampled service time indicates that timely completion is no longer feasible. In the implementation, these two mechanisms contribute to one aggregate deadline-drop counter used in reporting. Accordingly, B_s is the effective queue-loss count, $B_{\text{cap},s}$ is the structural finite-capacity loss count, and L_s^{dead} is the aggregate deadline-induced loss count.

At each PRB-control update, the simulator records the current URLLC cumulative mean sojourn time and queue occupancy. The controller then uses the most recent $T_M = 10$ s of these sampled URLLC values when forming its short-horizon URLLC congestion estimate. For eMBB and Safety, the implemented controller uses the current cumulative mean sojourn time of departed packets together with instantaneous queue occupancy. The averaging rule is therefore slice-dependent in the controller.

Completed-packet timely completion and URLLC tail metric. The completed-packet conditional timely-completion ratio is

$$R_{\text{cond},s} = \frac{N_s^{\text{ok}}}{N_s^{\text{dep}}}, \qquad N_s^{\text{dep}} > 0. \tag{13}$$

This metric is conditioned on completed packets and therefore excludes blocked, dropped, and unfinished packets. If $N_s^{\text{dep}} = 0$, it is left undefined.

For URLLC tail-delay tracking, the simulator maintains the generic sliding-window estimator

$$\hat{p}_{99}(t; T_w) = \text{quantile}_{0.99}\big(\{\, W_{n,U} : n \in \mathcal{D}_U(t - T_w, t) \,\}\big), \tag{14}$$

where $\mathcal{D}_U(t - T_w, t)$ is the set of URLLC packets departing within the window $(t - T_w, t]$.

Table 4
PRB-control, service-parallelism, and Safety PC5-stage delivery notation.

Symbol	Meaning
C_{tot}	Total PRB budget available at each gNB.
$n_s(t)$	Integer number of PRBs allocated to slice s at time t.
p_s	Number of PRBs required to activate one effective packet server for slice s.
$c_s(t) = \max(1, \lfloor n_s(t)/p_s \rfloor)$	PRB-induced number of effective packet servers in slice s.
$\mu_s^{\text{agg}}(t)$	Aggregate service capacity of slice s, $\mu_s^{\text{agg}}(t) = c_s(t)\mu_{0,s}$.
$M_s(t)$	Congestion score of slice s. In the implemented controller, URLLC includes sampled delay, queue occupancy, the controller-side tail estimate, and unfinished-arrival pressure, while Safety additionally includes queue/tail stress terms related to unfinished workload.
$\Psi_U(t), \Psi_S(t)$	Additional unfinished-work stress terms used in the URLLC and Safety congestion scores.
$f_s^\star(t)$	Instantaneous target PRB fraction derived from congestion scores.
$f_s(t)$	Smoothed PRB fraction applied to slice s.
$f_s^{\min}, f_s^{\max}$	Minimum and maximum PRB-fraction bounds for slice s.
α, β	Weights used in the congestion score for delay and queue occupancy.
ϵ	Small positive constant used to avoid division by zero or degenerate score normalization.
η	Smoothing factor for PRB-fraction updates.
T_M	Measurement window used for sampled short-horizon URLLC mean-sojourn and queue-occupancy averages. In the simulations, $T_M = 10$ s.
T_w^{ctrl}	Controller-side URLLC tail-delay window used for rapid tail tracking and emergency detection. In the simulations, $T_w^{\text{ctrl}} = 5$ s.
D_U^{emg}	Early-warning URLLC emergency threshold applied to the controller-side tail estimate $\hat{p}_{99,\text{ctrl}}(t)$.
$\dot{p}_{\text{emg}}$	Emergency threshold for the short-term increasing trend $\dot{p}_{99}(t)$ of the controller-side URLLC tail estimate.
$\mathbf{f}^{\text{emg}}$	Emergency PRB split applied when URLLC tail-delay risk is detected.
PC5	Direct sidelink interface used for Safety-side V2X delivery abstraction.
HARQ	Hybrid automatic repeat request mechanism included in the Safety-side PC5 delivery abstraction.
$T_{\text{HARQ}}, A_{\max}$	HARQ round-trip time and maximum number of allowed HARQ attempts.
a, γ_0	Slope and midpoint parameters of the logistic BLER mapping.
$R_{\text{e2e},S}$	Queue-side finite-horizon timely-completion ratio of the Safety slice.
$R_{\text{PC5}}^{(1)}$	First-attempt PC5 delivery reliability for Safety packets entering the downstream sidelink stage.
$R_{\text{PC5}}^{\text{HARQ}}$	HARQ-assisted PC5 delivery reliability after the allowed HARQ attempts.
Δ_{HARQ}	HARQ gain at the PC5 stage, $\Delta_{\text{HARQ}} = R_{\text{PC5}}^{\text{HARQ}} - R_{\text{PC5}}^{(1)}$.

The controller-side and reporting-window estimates are then defined as

$$\hat{p}_{99,\text{ctrl}}(t) = \hat{p}_{99}(t; T_w^{\text{ctrl}}), \tag{15}$$

$$\hat{p}_{99,\text{rep}}(t) = \hat{p}_{99}(t; T_w^{\text{rep}}). \tag{16}$$

In the simulations, $T_w^{\text{ctrl}} = 5$ s is used for PRB control, emergency detection, and admission-related safeguards, whereas $T_w^{\text{rep}} = 10$ s is used for plotted URLLC tail-delay traces. The estimate is left undefined when the corresponding window contains too few departed URLLC packets for stable high-quantile estimation.

Queue-side finite-horizon timely completion and loss decomposition. The queue-side finite-horizon timely completion ratio is

$$R_{\text{e2e},s} = \frac{N_s^{\text{ok}}}{N_s^{\text{arr}}}, \qquad N_s^{\text{arr}} > 0. \tag{17}$$

Here, N_s^{arr} is the total number of arrivals observed during the simulation horizon. Packets that are blocked, dropped by deadline enforcement, served after their deadline, or still unfinished at the end of the horizon all reduce $R_{\text{e2e},s}$.

The effective queue-loss fraction is

$$P_{\text{eff},s} = \frac{B_s}{N_s^{\text{arr}}}, \tag{18}$$

and its capacity-induced and deadline-induced components are

$$P_{\text{cap},s} = \frac{B_{\text{cap},s}}{N_s^{\text{arr}}}, \qquad P_{\text{dead},s} = \frac{L_s^{\text{dead}}}{N_s^{\text{arr}}}. \tag{19}$$

Therefore,

$$P_{\text{eff},s} = P_{\text{cap},s} + P_{\text{dead},s}. \tag{20}$$

The late-served and unfinished-by-horizon fractions are

$$P_{\text{late},s} = \frac{N_s^{\text{dep}} - N_s^{\text{ok}}}{N_s^{\text{arr}}}, \qquad P_{\text{pend},s} = \frac{N_s^{\text{arr}} - B_s - N_s^{\text{dep}}}{N_s^{\text{arr}}}, \tag{21}$$

where $P_{\text{late},s}$ counts departed packets that missed their deadlines and $P_{\text{pend},s}$ counts packets that were neither lost nor departed by the end of the horizon.

The complete finite-horizon decomposition is

$$1 = R_{\text{e2e},s} + P_{\text{cap},s} + P_{\text{dead},s} + P_{\text{late},s} + P_{\text{pend},s}. \tag{22}$$

The corresponding non-timely-completion fraction is

$$\text{Tail}_s = 1 - R_{\text{e2e},s}. \tag{23}$$

Safety PC5-stage delivery metrics. For the downstream Safety PC5 sidelink stage, the simulator reports link-attempt-based delivery metrics separately from the gNB queue-side metrics. Let $N_{\text{PC5}}^{\text{link}}$ denote the number of first PC5 link attempts, $Y_{\text{PC5}}^{(1)}$ the number of first-attempt successes, and $Y_{\text{PC5}}^{\text{HARQ}}$ the number of links that are eventually successful within the allowed HARQ attempts. The first-attempt and HARQ-assisted PC5 delivery reliabilities are

$$R_{\text{PC5}}^{(1)} = \frac{Y_{\text{PC5}}^{(1)}}{N_{\text{PC5}}^{\text{link}}}, \qquad R_{\text{PC5}}^{\text{HARQ}} = \frac{Y_{\text{PC5}}^{\text{HARQ}}}{N_{\text{PC5}}^{\text{link}}}, \tag{24}$$

and the HARQ gain is

$$\Delta_{\text{HARQ}} = R_{\text{PC5}}^{\text{HARQ}} - R_{\text{PC5}}^{(1)}. \tag{25}$$

These PC5 reliabilities describe the downstream sidelink delivery stage and are not part of the additive queue-side decomposition in (22). The simulator also maintains separate queue-side counts for the per-link PC5 queues, but those are reported independently from the main gNB slice-queue metrics.

SLA targets and interpretation. The slice-level latency targets follow 3GPP service categories and latency regimes commonly used in 5G and NR-V2X studies [2,3,15]. In the representative scenario, the slice deadlines are

$$D_U = 1 \text{ ms}, \qquad D_S = 20 \text{ ms}, \qquad D_E = 50 \text{ ms}. \tag{26}$$

These targets serve as slice-level benchmarks for queueing, scheduling, and PRB-control behavior. The reported URLLC 1 ms result is therefore evaluated through the gNB slice-queue sojourn-time target and the

Algorithm 1: Deadline-aware finite-capacity slice queue with EDF ordering, queue expiration, and predictive feasibility dropping.

Input : event at time t for slice s; queue state $(Q_s, \mathcal{W}_s, c_s, K_s)$; deadline D_s; service sampler G_{k_s}
Output : updated queue state, departure events, and queue-side counters

if *the event is an arrival* **then**
  increase the total-arrival counter
  update the queue-area integral
  purge expired waiting packets if queue-expiration is enabled
  if $Q_s(t) \geq K_s(t)$ **then**
    increase the finite-capacity blocking counters and return
  if $Q_s(t) < c_s(t)$ **then**
    sample $S \sim G_{k_s}$
    if *predictive feasibility is enabled* ***and*** $t + S > t + D_s$ **then**
      increase the aggregate blocked/deadline-drop counters and return
    schedule a departure at $t + S$
    increase $Q_s(t)$ and return
  append the packet $(A = t, \Theta = t + D_s)$ to the waiting queue $\mathcal{W}_s$
  if *EDF is enabled* **then**
    sort $\mathcal{W}_s$ by increasing deadline
  increase $Q_s(t)$
else
  remove the earliest departure event
  record the sojourn time $W_{n,s} = t - A_{n,s}$ for later SLA aggregation
  increase the departure counter and decrease $Q_s(t)$
  purge expired waiting packets if queue-expiration is enabled
  while $\mathcal{W}_s$ *is not empty* **do**
    remove the head-of-line packet (A, Θ)
    sample $S \sim G_{k_s}$
    if *predictive feasibility is enabled* ***and*** $t + S > \Theta$ **then**
      increase the aggregate blocked/deadline-drop counters
      decrease $Q_s(t)$
    else
      schedule a departure at $t + S$ and stop
  update the next-departure pointer

reported URLLC tail metric p_{99}. Lower-layer coding, link adaptation, retransmission timing, radio calibration, and other full protocol-stack effects remain outside the present slice-level abstraction.

5. gNB model and dynamic PRB control

Each gNB manages three logical slices, URLLC (U), eMBB (E), and Safety (S), under a fixed physical resource block (PRB) budget C_{tot}. The controller periodically updates the slice PRB split and thereby changes the number of effective packet servers available to each slice. In the implementation, the PRB-control update period is 10 ms. The control logic is mainly local to each gNB, while limited cross-gNB awareness is retained through exchanged URLLC tail-delay and emergency-state indicators.

5.1. Congestion observation and score construction

The controller is designed so that URLLC control reacts to tail-latency risk rather than to backlog alone, PRB reallocation acts directly on packet-level service parallelism through the number of effective packet servers, and allocation updates remain stable under bursty traffic through score clipping, smoothing, emergency override, and conservative recovery.

At each control instant, the gNB updates recent queueing measurements and computes slice congestion scores from slice-specific delay, occupancy, tail, and unfinished-work signals. For URLLC, the controller uses a 10 s sampled mean-sojourn and queue-occupancy window, the controller-side 99th-percentile tail-delay estimate $\hat{p}_{99,\text{ctrl}}(t)$ computed over a 5 s window, and an unfinished-arrival pressure term.

The URLLC score is

$$M_U(t) = \alpha \frac{\overline{W}_U(t)}{D_U + \varepsilon} + \beta \frac{\overline{Q}_U(t)}{K_U(t)} + \max\left(0, \frac{\hat{p}_{99,\text{ctrl}}(t)}{D_U} - 1\right) + \Psi_U(t), \tag{27}$$

where $M_U(t)$ is the URLLC congestion score, $\overline{W}_U(t)$ is the recent mean URLLC sojourn time, $\overline{Q}_U(t)$ is the recent URLLC queue occupancy, D_U is the URLLC deadline, and $K_U(t)$ is the current URLLC queue capacity. The coefficients α and β weight delay and occupancy, $\varepsilon > 0$ is a small numerical constant, and $\Psi_U(t)$ is an additional implementation-level stress term that increases when URLLC departures lag persistently behind arrivals.

For eMBB and Safety, the main score components are delay and occupancy:

$$M_E(t) = \alpha \frac{\overline{W}_E(t)}{D_E + \varepsilon} + \beta \frac{Q_E(t)}{K_E(t)}, \tag{28}$$

$$M_S(t) = \alpha \frac{\overline{W}_S(t)}{D_S + \varepsilon} + \beta \frac{Q_S(t)}{K_S(t)} + \Psi_S(t), \tag{29}$$

where $M_E(t)$ and $M_S(t)$ denote the eMBB and Safety congestion scores, respectively. In (28), $\overline{W}_E(t)$, $Q_E(t)$, D_E, and $K_E(t)$ are the eMBB mean sojourn time, current queue occupancy, deadline, and queue capacity. In (29), $\overline{W}_S(t)$, $Q_S(t)$, D_S, and $K_S(t)$ are the corresponding Safety quantities, while $\Psi_S(t)$ is an additional Safety-stress term that increases when unfinished Safety traffic accumulates. In the implementation, the slice scores are additionally clipped so that short transients do not monopolize the PRB budget.

5.2. PRB allocation rule

At each control instant, the controller first computes the slice congestion scores $M_U(t)$, $M_E(t)$, and $M_S(t)$. These scores summarize recent slice queueing conditions, while the URLLC score additionally captures controller-side tail-latency pressure through $\hat{p}_{99,\text{ctrl}}(t)$ and unfinished-arrival pressure. A larger score therefore indicates greater short-horizon urgency for PRB assignment.

The score vector is converted into a target PRB-fraction vector according to

$$f_s^\star(t) = \frac{M_s(t) + \varepsilon}{\sum_{r \in S} \left(M_r(t) + \varepsilon\right)}, \qquad s \in S = \{U, E, S\}, \tag{30}$$

where $f_s^\star(t)$ is the target PRB fraction of slice s, $M_s(t)$ is the corresponding slice score, and $\varepsilon > 0$ is a small constant that avoids numerical degeneracy when all scores are close to zero. The resulting fractions are clipped to admissible bounds and renormalized.

To reduce oscillations, the target fractions are smoothed with the previously applied allocation:

$$f_s(t) = \eta f_s(t^-) + (1 - \eta) f_s^\star(t), \qquad 0 < \eta < 1, \tag{31}$$

where $f_s(t)$ is the applied PRB fraction after smoothing, $f_s(t^-)$ is the fraction used immediately before time t, and η is the smoothing parameter.

If the URLLC tail enters the emergency region, the score-based allocation is temporarily overridden by the fixed emergency split

$$\mathbf{f}^{\text{emg}} = (0.90,\ 0.07,\ 0.03), \tag{32}$$

whose entries correspond to the URLLC, eMBB, and Safety PRB fractions. Under this mode, most PRBs are immediately redirected to URLLC. After

the emergency condition disappears, the controller returns gradually to the score-driven allocation through the smoothing rule in (31).

The smoothed slice fractions are then converted into integer PRB allocations:

$$n_s(t) \approx \text{round}\big(C_{\text{tot}} f_s(t)\big), \qquad s \in S, \tag{33}$$

where $n_s(t)$ is the number of PRBs assigned to slice s and C_{tot} is the total PRB budget of the gNB. A subsequent correction step enforces the exact PRB budget and slice-specific minimum safeguards. The implemented controller is therefore a heuristic score-driven allocation rule with emergency override and bounded post-correction.

The final PRB allocation determines the number of effective packet servers:

$$c_s(t) = \max\left(1, \left\lfloor \frac{n_s(t)}{p_s} \right\rfloor\right), \tag{34}$$

where $c_s(t)$ is the number of effective packet servers assigned to slice s and p_s is the PRB requirement of one packet server for that slice. The corresponding aggregate service capacity is

$$\mu_s^{\text{agg}}(t) = c_s(t)\mu_{0,s}, \tag{35}$$

with $\mu_{0,s}$ denoting the fixed service rate of one packet server in slice s. PRB allocation therefore changes queue behavior by changing packet-level service parallelism.

5.3. Emergency logic and conservative recovery

The controller includes an emergency mode to protect the URLLC tail before the 1 ms target is persistently exceeded. A 5 s window is used to compute the controller-side URLLC tail estimate $\hat{p}_{99,\text{ctrl}}(t)$, and a short history of the most recent controller-side samples is used to approximate its local slope $\dot{p}_{99}(t)$.

Emergency mode is activated when either the controller-side URLLC tail estimate exceeds an early-warning threshold or its short-term trend becomes too steep:

$$\hat{p}_{99,\text{ctrl}}(t) > D_U^{\text{emg}} \quad \text{or} \quad \dot{p}_{99}(t) > \dot{p}_{\text{emg}}. \tag{36}$$

Here, D_U^{emg} is the emergency threshold for the URLLC 99th-percentile delay and $\dot{p}_{\text{emg}}$ is the threshold for its short-term slope. In the implementation,

$$D_U^{\text{emg}} = 0.95 \text{ ms}, \qquad \dot{p}_{\text{emg}} = 0.5 \text{ ms/s}. \tag{37}$$

When emergency mode is active, the controller applies the URLLC-dominant split

$$\mathbf{f}^{\text{emg}} = (0.90,\ 0.07,\ 0.03). \tag{38}$$

Recovery is intentionally conservative. In the implementation, the controller exits emergency mode only after the URLLC tail remains below 0.8 ms for more than 3 s. PRBs are then released gradually before the standard smoothing rule is fully resumed. This prevents abrupt PRB withdrawal immediately after a transient URLLC stress period.

5.4. Admission-control coupling

In this paper, deadline-aware protection is implemented in two stages: an arrival-time URLLC admission guard and a predictive service-start feasibility check.

The queueing system always enforces the structural capacity condition

$$Q_s(t) < K_s(t). \tag{39}$$

Here, $Q_s(t)$ is the current number of packets in slice s, while $K_s(t)$ is the total instantaneous capacity of that slice, including both in-service and waiting packets.

In addition to this structural condition, the proposed controller applies an extra URLLC admission guard:

$$G_U(t) = \big[Q_U(t) \le 0.85 K_U(t)\big] \ \vee \ \big[\hat{p}_{99,\text{ctrl}}(t) \le 1.2 \text{ ms}\big]. \tag{40}$$

The guard allows admission when either the URLLC queue occupancy remains below 85% of the current URLLC capacity or the URLLC tail estimate remains below 1.2 ms.

If this guard fails at the serving gNB, the implementation attempts lightweight redirection to a neighboring gNB that can still admit new URLLC traffic, preferring the neighbor with the smaller current URLLC load ratio. After admission, deadline feasibility is checked again when service is about to begin. If the sampled service requirement implies that timely completion is no longer feasible, the packet is dropped before consuming service time. This predictive deadline drop remains one of the main queue-level protection mechanisms of the proposed controller.

5.5. Implemented tail-guided PRB controller

Algorithm 2 summarizes the implemented mission-critical controller executed at each gNB every 10 ms.

6. Channel, mobility, and PC5 sidelink model

This section defines the radio, mobility, and Safety-side PC5 abstractions that connect vehicular movement and link variation to the slice queues. The model generates time-varying signal-to-interference-plus-noise ratio (SINR), block error rate (BLER), and HARQ outcomes that influence packet-level service and delivery behavior in the discrete-event simulation.

All traffic first enters the gNB-side Uu slice queues. URLLC and eMBB packets are evaluated at the gNB slice-queue level, with lower-layer radio effects absorbed into service-time variability and PRB requirements. Safety packets additionally pass through a downstream PC5 sidelink/HARQ delivery stage after gNB queueing. Queue-side finite-horizon timely-completion metrics and PC5-stage delivery metrics are reported separately so that gNB queue loss, unfinished backlog, and downstream delivery effects remain clearly distinguished.

6.1. Noise, path loss, and shadowing

At carrier frequency $f_c = 5.9$ GHz and bandwidth B, the receiver noise power is modeled as

$$N(B) = N_0 F B, \tag{41}$$

where N_0 is the thermal noise density and

$$F = 10^{NF/10} \tag{42}$$

is the receiver noise figure in linear scale, with NF expressed in dB.

Large-scale attenuation is modeled using a dual-slope log-distance path-loss law:

$$\text{PL}_{\text{det}}(d) = 32.4 + 10n(d)\log_{10}(d) + 20\log_{10}\left(\frac{f_c}{1 \text{ GHz}}\right), \tag{43}$$

where d is the transmitter–receiver distance in meters. The distance-dependent path-loss exponent is

$$n(d) = \begin{cases} 2, & d < 50 \text{ m}, \\ 3, & d \ge 50 \text{ m}. \end{cases} \tag{44}$$

Lognormal shadowing is included as

$$X_\sigma \sim \mathcal{N}(0, \sigma_{\text{sh}}^2), \tag{45}$$

so that the total path loss in dB becomes

$$\text{PL}_{\text{dB}}(d) = \text{PL}_{\text{det}}(d) + X_\sigma. \tag{46}$$

The corresponding linear attenuation factor is

$$\text{PL}_{\text{lin}}(d) = 10^{\text{PL}_{\text{dB}}(d)/10}. \tag{47}$$

Algorithm 2: Implemented tail-guided PRB control and URLLC admission at gNB k.

Input : current time t; local slice states; previous PRB fractions $\mathbf{f}(t^-)$; network reports netMetrics
Output : integer PRB allocation $\mathbf{n}(t)$ and effective server counts $\mathbf{c}(t)$

compute $\hat{p}_{99,\mathrm{ctrl}}(t)$ from URLLC departures in the last 5 s
update the recent controller-side tail history and estimate its slope $\dot{p}_{99}(t)$
compute URLLC short-horizon mean sojourn and mean occupancy over the last 10 s
compute cumulative mean sojourn and instantaneous occupancy for eMBB and Safety
compute unfinished-arrival pressures for URLLC and Safety, and set the Safety-stress flag
form $M_U(t)$ from normalized URLLC delay, normalized URLLC occupancy, tail excess, and unfinished-arrival pressure
form $M_E(t)$ from normalized eMBB delay and occupancy
form $M_S(t)$ from normalized Safety delay and occupancy, plus Safety tail/stress penalties
clip all scores and convert them into a normalized target fraction vector $\mathbf{f}^\star(t)$
if *another gNB is already in emergency mode* **then**
    force a larger local URLLC share
if *the worst gNB has high URLLC tail delay and another gNB remains well below the target* **then**
    nudge the worst gNB upward and the best gNB downward in URLLC share
if $\hat{p}_{99,\mathrm{ctrl}}(t) > 0.95$ *ms* ***or*** $\dot{p}_{99}(t) > 0.5$ *ms/s* **then**
    enter emergency mode and set $\mathbf{f}^\star(t) \leftarrow (0.90, 0.07, 0.03)$
else if *the controller is already in emergency mode* **then**
    if $\hat{p}_{99,\mathrm{ctrl}}(t) < 0.80$ *ms for more than* 3 *s* **then**
        leave emergency mode and start conservative recovery
    else
        keep $\mathbf{f}^\star(t) \leftarrow (0.90, 0.07, 0.03)$
if *the controller is in recovery mode* **then**
    blend the previous fractions with the score-driven fractions
apply smoothing: $\mathbf{f}(t) = 0.5\,\mathbf{f}(t^-) + 0.5\,\mathbf{f}^\star(t)$
clip $\mathbf{f}(t)$ to admissible bounds and renormalize
round $\mathbf{f}(t)$ to integer PRBs $\mathbf{n}(t)$
enforce minimum PRB safeguards for URLLC, eMBB, and Safety; if Safety stress is active, raise the Safety/eMBB floors
if *URLLC is safely below the margin and under-utilized* **then**
    release at most one URLLC PRB to Safety or eMBB if their safeguard floor is still unmet
map PRBs to effective servers using $c_s(t) = \max\left(1, \lfloor n_s(t)/p_s \rfloor\right)$
update slice capacities as $K_s(t) = c_s(t) + W_s$
`/* URLLC arrival handling at time t */`
if $t < 2$ *s* ***or*** $Q_U(t) \le 0.85 K_U(t)$ ***or*** $\hat{p}_{99,\mathrm{ctrl}}(t) \le 1.2$ *ms* **then**
    admit the packet to the local URLLC queue
else if *a neighboring gNB can admit new URLLC traffic* **then**
    redirect the packet to the admissible neighbor with the lowest URLLC load ratio
else
    do not admit the packet locally

6.2. Rayleigh fading and SINR model

Small-scale fading is modeled using Rayleigh fading:

$$h \sim \mathcal{CN}(0,1), \qquad H = |h|^2. \tag{48}$$

Thus, the channel power gain H follows an exponential distribution with unit mean. Rayleigh fading is a widely used stochastic model for fast small-scale fading in mobile wireless systems [60–62]. In this work, it is adopted as a lightweight abstraction for generating time-varying SINR, BLER, and HARQ outcomes within the slice-level queueing and PRB-control simulator. The objective is not geometry-specific link-level calibration, but the evaluation of deadline-aware scheduling and slicing policies under a common source of radio variability. This choice keeps the simulator computationally efficient, exposes the controller to rapid channel fluctuations, and ensures that all compared policies experience the same fading and interference assumptions. More detailed channel models can be substituted without changing the queueing, deadline-handling, or PRB-control formulation.

For a transmitter with power P_{tx}, transmit antenna gain G_{tx}, receiver antenna gain G_{rx}, and link distance d, the received signal power is

$$S = \frac{P_{\mathrm{tx}} G_{\mathrm{tx}} G_{\mathrm{rx}} H}{\mathrm{PL}_{\mathrm{lin}}(d)}. \tag{49}$$

Let $\mathcal{I}$ denote the set of simultaneous interferers on the same resource. The aggregate interference is

$$I = \sum_{i \in \mathcal{I}} \frac{P_i G_{i,\mathrm{tx}} G_{\mathrm{rx}} H_i}{\mathrm{PL}_{i,\mathrm{lin}}(d_i)}, \tag{50}$$

where P_i, H_i, and d_i are the transmit power, Rayleigh channel power gain, and distance of interferer i, respectively. The instantaneous SINR over bandwidth B is then

$$\gamma = \frac{S}{I + N(B)}. \tag{51}$$

These SINR values drive the BLER mapping, HARQ retransmission decisions, and Safety-side PC5 delivery outcomes.

6.3. BLER and decoding outcome

Packet decoding is modeled through an SINR-dependent BLER curve. In the simulator, the BLER is represented by a smooth logistic approximation:

$$\mathrm{BLER}(\gamma) = \frac{1}{1 + \exp\left[a\left(\gamma_{\mathrm{dB}} - \gamma_0\right)\right]}, \tag{52}$$

where $a > 0$ controls the transition steepness, γ_0 is the midpoint SINR threshold, and

$$\gamma_{\mathrm{dB}} = 10 \log_{10}(\gamma). \tag{53}$$

The success probability of one transmission attempt is therefore

$$P_{\mathrm{succ}}(\gamma) = 1 - \mathrm{BLER}(\gamma). \tag{54}$$

This logistic mapping preserves the monotonic relation between SINR and decoding success while remaining lightweight enough for event-level simulation.

6.4. Mobility and gNB association

Each UE is modeled as a mobile vehicle with position

$$\mathbf{x}_m(t) \in \mathbb{R}^2 \tag{55}$$

and mobility speed v_m. Mobility follows a waypoint model. If $\mathbf{w}_m$ is the current waypoint of UE m, then the position update over time step Δt is

$$\mathbf{x}_m(t + \Delta t) = \mathbf{x}_m(t) + v_m \frac{\mathbf{w}_m - \mathbf{x}_m(t)}{\|\mathbf{w}_m - \mathbf{x}_m(t)\|} \Delta t. \tag{56}$$

When the UE reaches the neighborhood of its current waypoint, a new waypoint is sampled uniformly over the deployment area. This mobility process continuously changes path loss, fading, interference geometry, and serving-gNB association.

The serving gNB is selected according to the nearest-gNB rule:

$$g_m(t) = \arg\min_j \left\| \mathbf{x}_m(t) - \mathbf{x}_j^{\mathrm{gNB}} \right\|, \tag{57}$$

where $\mathbf{x}_j^{\mathrm{gNB}}$ is the position of gNB j. A handover occurs when the nearest gNB changes. In the queueing model, this association determines the gNB queue that receives newly generated Uu packets.

6.5. Slice identity and traffic generation

Each UE belongs to one slice $s \in S = \{U, E, S\}$. Slice identity determines packet size L_s, deadline D_s, traffic intensity, and delivery path. URLLC UEs generate small latency-critical packets, eMBB UEs generate larger throughput-oriented packets, and Safety UEs generate event-driven vehicular safety messages.

Traffic generation follows a time-varying Poisson model. For slice s, the instantaneous arrival rate is

$$\lambda_s(t) = \lambda_s^{(0)} + a_s \sin(\omega_s t) + \lambda_s^{(\text{burst})}(t), \tag{58}$$

where $\lambda_s^{(0)}$ is the nominal slice arrival rate, a_s is the oscillation amplitude in packets/s, ω_s controls the slow load-variation period, and $\lambda_s^{(\text{burst})}(t)$ represents short burst events. The sinusoidal term captures gradual traffic variation, while the burst term captures sudden URLLC or Safety stress periods. Interarrival times are exponentially distributed with the current rate $\lambda_s(t)$, so the traffic model remains consistent with the $M(t)$ arrival process used in the slice queues.

6.6. PC5 sidelink resource state for safety traffic

After the gNB queueing stage, Safety traffic enters a PC5 sidelink abstraction. Each Safety UE maintains a PC5 state vector

$$\left(r_{\text{PC5}}, t_{\text{exp}}, T_{\text{res}}, P_{\text{tx}}, G_{\text{ant}}\right), \tag{59}$$

where r_{PC5} is the reserved sidelink resource index, t_{exp} is the reservation expiry time, T_{res} is the reservation interval, P_{tx} is the transmit power, and G_{ant} is the antenna gain.

A reservation is valid for $t < t_{\text{exp}}$. When the reservation expires, the Safety UE selects a new resource using a Mode-2-like autonomous PC5 reservation rule. Let $\mathcal{R}_{\text{PC5}}$ denote the set of available PC5 resources. A resource is selected as

$$r_{\text{PC5}}(t) \in \mathcal{R}_{\text{PC5}}, \tag{60}$$

with the implementation avoiding resources already sensed as occupied whenever such information is available.

The set of neighbors of Safety UE m is

$$\mathcal{N}_m(t) = \left\{\ell : \left\|\mathbf{x}_\ell(t) - \mathbf{x}_m(t)\right\| \le R_{\text{PC5}}\right\}, \tag{61}$$

where R_{PC5} is the sidelink communication range. Interference on the PC5 link is created by other Safety UEs transmitting on the same PC5 resource.

6.7. Safety PC5/HARQ delivery abstraction

After a Safety packet departs successfully from the gNB queue, the simulator evaluates its PC5 sidelink delivery. For Safety packet n, the first PC5 transmission attempt occurs at time $t_n^{(1)}$. If decoding fails, retransmission attempts are scheduled according to the HARQ round-trip time T_{HARQ}, up to a maximum of $A_{\max}$ attempts. The time of attempt a is

$$t_n^{(a)} = t_n^{(1)} + (a-1)T_{\text{HARQ}}, \qquad a = 1, \ldots, A_{\max}. \tag{62}$$

For each attempt, the SINR $\gamma_n^{(a)}$ is computed using (51), and the decoding failure probability is obtained from (52). Let $U_n^{(a)} \sim \mathcal{U}(0,1)$. Attempt a succeeds if

$$U_n^{(a)} > \text{BLER}\left(\gamma_n^{(a)}\right). \tag{63}$$

The Safety-side delivery indicator is then

$$H_n = \mathbb{1}\left[\exists a \le A_{\max} : U_n^{(a)} > \text{BLER}\left(\gamma_n^{(a)}\right),\ t_n^{(a)} \le \Theta_n\right], \tag{64}$$

where $\Theta_n = A_n + D_S$ is the absolute Safety deadline. A Safety packet is counted as successfully delivered only if it satisfies the gNB queueing deadline and at least one PC5/HARQ attempt succeeds before the packet deadline.

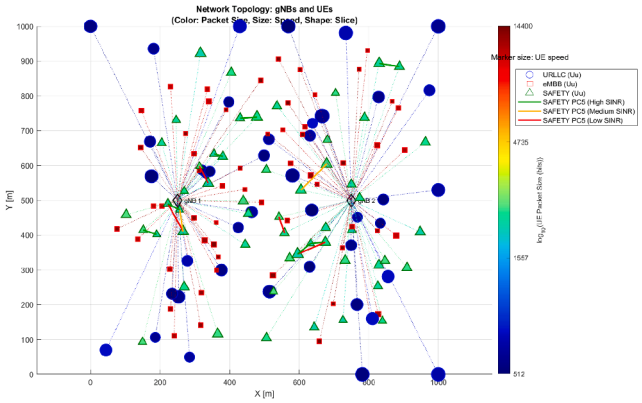


Fig. 2. NR-V2X topology with two gNBs and single-slice UEs. Marker shape denotes slice type, marker color denotes UE packet size, and marker size denotes UE speed. Dotted lines indicate Uu gNB associations, while solid safety links show representative PC5 sidelink connections under high-, medium-, and low-SINR conditions.

This delivery indicator is used to compute the separate PC5-stage metrics defined in Section 4.6. These downstream metrics are reported separately from the queue-side finite-horizon R_{e2e}, so that PC5/HARQ delivery effects are not mixed with gNB queue loss, late service, or unfinished backlog.

The radio and mobility model influences the slice queues in three ways. First, mobility and association determine which gNB receives each newly generated packet. Second, path loss, fading, and interference generate time-varying SINR values that shape Safety PC5/HARQ delivery outcomes and PC5-stage delivery reliability. Third, the resulting completions, retransmissions, and failures update the queue-side timely-completion, loss-decomposition, and PC5-stage metrics used in the results section.

7. Results

7.1. Simulation framework and experimental setup

The proposed framework was implemented in MATLAB as a custom packet-level causal discrete-event simulator. The simulator combines event-driven slice arrivals and departures with periodic mobility, traffic, channel, PRB-control, and PC5 scheduling updates. Unless otherwise stated, all compared policies are evaluated under the same network geometry, mobility model, traffic-generation process, channel abstraction, queueing structure, and performance-metric definitions; only the slice-control logic is changed across the proposed method, baselines, and ablation variants.

The representative scenario consists of two gNBs serving vehicle-mounted UEs over a 1000×1000 m area. The main experimental, queueing, and controller parameters are summarized in Table 5. Randomized experiments are repeated over the number of independent seeds stated in the corresponding result tables, and the same evaluation workflow is used across all configurations.

Fig. 2 illustrates a representative network snapshot. Two gNBs and all active UEs in the 1000×1000 m^2 area are shown. Marker shape encodes slice identity: circle for URLLC, square for eMBB, and triangle for Safety. Marker size reflects UE speed, and marker colour encodes packet size. Dotted lines indicate UE–gNB associations, while coloured PC5 arrows depict high-, medium-, and low-SINR sidelinks. The resulting scenario is heterogeneous, mobility-driven, and interference-limited, as expected in representative vehicular deployments.

Table 5
Default simulation framework and experimental setup.

Parameter	Value
Simulation duration / random seed	250 s / 42
Area / gNBs / inter-gNB distance	1000×1000 m / 2 / 500 m
Slices	URLLC (U), eMBB (E), Safety (S)
Mobility / mobility update	random waypoint / 1 s
Initial UEs / new UE process	5 per slice per gNB / 5 new UEs every 30 s
Base per-gNB arrival rates	$\lambda_U^{(0)} = 180,\ \lambda_E^{(0)} = 520,\ \lambda_S^{(0)} = 120$ pkt/s
Traffic variability	$a_s = 5$ pkt/s, burst multiplier = 1.5, burst period = 100 s, burst duration = 10 s
NR timing / PRBs per gNB	120 kHz SCS, 0.125 ms slot / $C_{\text{tot}} = 100$
PRBs per packet server	$p_U = 6,\ p_E = 3,\ p_S = 8$
Initial PRB split / effective servers	$(0.60, 0.25, 0.15)$ / $(10, 8, 1)$ for (U, E, S)
Service-time shape	$k_U = 4,\ k_E = 1,\ k_S = 1$
Per-server service rates	$\mu_{0,U} \approx 2666.7,\ \mu_{0,E} \approx 533.3,\ \mu_{0,S} \approx 888.9$ pkt/s
Waiting capacity / deadlines	$W_U = 10,\ W_E = 5,\ W_S = 5$ / $D_U = 1$ ms, $D_E = 50$ ms, $D_S = 20$ ms
Queue discipline / deadline handling	EDF, queue-expiry enabled, predictive feasibility drop enabled
Emergency split / thresholds	$(0.90, 0.07, 0.03)$, $D_U^{\text{emg}} = 0.95$ ms, $\dot{p}_{\text{emg}} = 0.5$ ms/s
Admission guard	$Q_U(t) \le 0.85 K_U(t)$ or $\hat{p}_{99,\text{ctrl}}(t) \le 1.2$ ms
Controller parameters	$\alpha = 0.6,\ \beta = 0.4,\ \eta = 0.50,\ f_s^{\min} = 0.06,\ f_s^{\max} = 0.90$
Update / window parameters	PRB = 10 ms, channel = 10 ms, PC5 = 100 ms; $T_w^{\text{ctrl}} = 5$ s, $T_M = 10$ s, $T_w^{\text{rep}} = 10$ s
Recovery rule	exit emergency after $\hat{p}_{99,\text{ctrl}}(t) < 0.8$ ms for 3 s; recovery blend 0.90/0.10
Channel abstraction	$f_c = 5.9$ GHz, $B_{\text{PRB}} = 180$ kHz, $NF = 7$ dB, $\sigma_{\text{sh}} = 4$ dB, $P_{\text{tx}} = 1$ W
PC5/HARQ parameters	10 resources, threshold = 5 dB, reservation = 0.1 s, PSCCH/PSSCH = 0.5/0.5 ms, $T_{\text{HARQ}} = 3$ ms, $A_{\max} = 8$, BLER $(a, \gamma_0) = (0.2, -5)$
Main URLLC target	reported URLLC $p_{99} \le 1$ ms at the gNB queueing/scheduling level

7.2. URLLC tail-delay performance

To plot the URLLC 99th-percentile sojourn-time traces, the simulator uses the reporting-window estimator $\hat{p}_{99,\text{rep}}(t)$, which is obtained from the generic sliding-window estimator in (14) by setting $T_w = T_w^{\text{rep}} = 10$ s. The PRB controller itself uses the shorter controller-side estimate $\hat{p}_{99,\text{ctrl}}(t)$ with $T_w^{\text{ctrl}} = 5$ s. This window length balances responsiveness and statistical stability: under the considered load conditions, each window contains on the order of 10^3 URLLC departures, which is sufficient for high-quantile estimation while allowing the controller to react to mobility-driven load shifts and bursty arrivals.

Fig. 3 shows the URLLC p_{99} sojourn-time traces for the two gNBs, computed using a 10 s sliding window. After the short initial transient, both traces remain below the 1 ms URLLC slice-level queueing target and fluctuate mainly around 0.84–0.91 ms. Residual oscillations are caused by traffic bursts, mobility-driven load variation, and transient PRB reallocations. The close alignment between the two traces indicates that the controller maintains comparable URLLC tail-delay behavior across the two gNBs.

The result also highlights the joint role of tail-guided PRB control and deadline-aware queue protection. The controller increases URLLC service parallelism when the controller-side tail estimate $\hat{p}_{99,\text{ctrl}}(t)$ approaches the early-warning region, while deadline-enforcement mechanisms prevent packets that are unlikely to complete on time from consuming scarce service capacity. Together, these mechanisms keep short congestion episodes from developing into persistent URLLC tail-delay excursions.

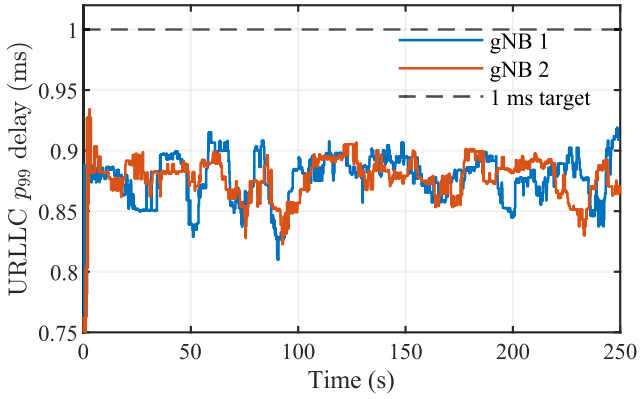


Fig. 3. URLLC p_{99} slice-queue sojourn delay for both gNBs using a 10 s sliding window. The dashed line denotes the 1 ms URLLC slice-level queueing target. After the warm-up phase, both gNBs remain below the target.

7.3. Finite-horizon timely-completion and loss decomposition

Table 6 reports the finite-horizon timely-completion and loss decomposition in the representative NR-V2X scenario. The reported decomposition follows

$$1 = R_{\text{e2e},s} + P_{\text{cap},s} + P_{\text{dead},s} + P_{\text{late},s} + P_{\text{pend},s}, \tag{65}$$

where $P_{\text{cap},s}$ is capacity-induced queue loss, $P_{\text{dead},s}$ is deadline-induced queue loss, $P_{\text{late},s}$ is the late-served fraction, and $P_{\text{pend},s}$ is the unfinished-by-horizon backlog fraction. The aggregate effective queue loss is

$$P_{\text{eff},s} = P_{\text{cap},s} + P_{\text{dead},s}. \tag{66}$$

Thus, $P_{\text{eff},s}$ is not an additional term in the additive decomposition; it is the sum of the capacity-induced and deadline-induced queue-loss components. In the representative run, $P_{\text{late},s} = 0$ for all three slices.

Table 6
Representative-scenario SLA and finite-horizon decomposition results.

Slice	$p_{99,s}$ (ms)	R_{e2e} (%)	P_{eff} (%)	P_{dead} (%)	P_{pend} (%)	R_{cond} (%)
URLLC	0.880	99.372	0.628	0.628	0.000	100.000
eMBB	8.657	99.999	0.000	0.000	0.0007	100.000
Safety	5.159	99.005	0.000	0.000	0.995	100.000

The URLLC slice achieves a p_{99} delay of 0.880 ms, remaining below the 1 ms slice-level queueing target. Its finite-horizon timely-completion ratio is $R_{\text{e2e},U} = 99.372\%$, with $P_{\text{eff},U} = 0.628\%$ effective queue loss. Since $P_{\text{cap},U} = 0$ and $P_{\text{dead},U} = 0.628\%$, the observed URLLC queue loss is entirely generated by the deadline-enforcement stage rather than by structural buffer overflow. This shows that the controller protects URLLC tail performance by removing packets that no longer fit the queue-level deadline budget instead of allowing them to consume service capacity and inflate the completed-packet tail delay.

The eMBB and Safety slices show no observed queue-level loss in this representative run. For eMBB, the finite-horizon gap is negligible, with only 0.0007% of packets pending at the observation horizon. For Safety, $R_{\text{e2e},S} = 99.005\%$ and $P_{\text{pend},S} = 0.995\%$, while $P_{\text{eff},S} = P_{\text{dead},S} = 0$.

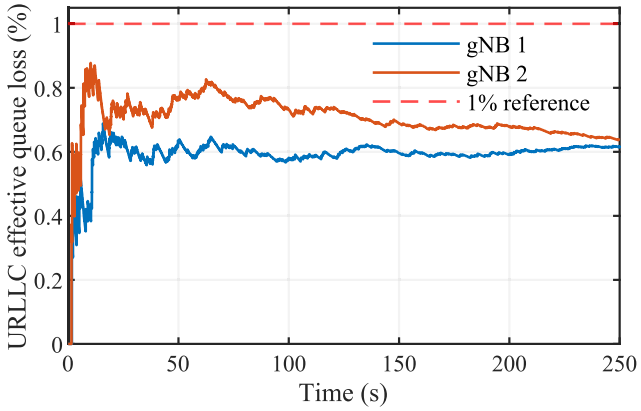


Fig. 4. URLLC effective queue loss $P_{\text{eff},U}$ for both gNBs. The dashed line denotes the 1% reference. The final values remain around 0.6%–0.7%; in the representative run, the URLLC effective queue loss is entirely deadline-induced.

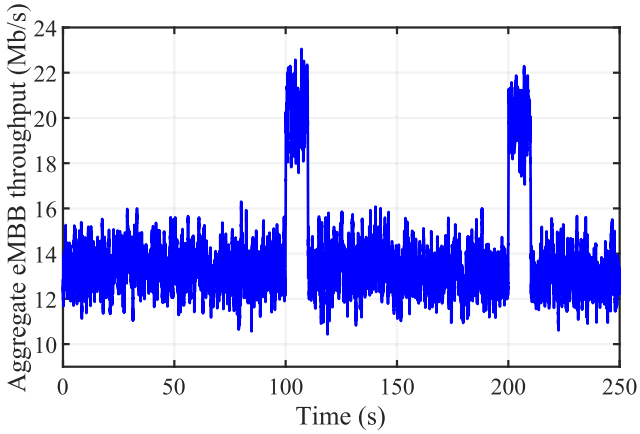


Fig. 5. Instantaneous aggregate eMBB delivered throughput across both gNBs. The eMBB slice sustains approximately 12–15 Mb/s under nominal operation and exhibits short peaks when residual PRB capacity becomes available.

Therefore, the Safety gap in this representative run should be interpreted as unfinished-by-horizon backlog, not as gNB queue blocking or deadline-induced queue loss.

7.4. URLLC effective queue loss

Fig. 4 reports the time evolution of URLLC effective queue loss for both gNBs. The plotted quantity corresponds to $P_{\text{eff},U}$. The dashed line denotes the 1% reference. After the initial transient, the URLLC effective queue loss stabilizes below this reference and remains around 0.6%–0.7%, consistently with Table 6.

The URLLC effective queue loss should therefore be interpreted as controlled deadline-aware filtering rather than uncontrolled congestion loss. In the representative run, it is entirely generated by the deadline-enforcement stage. Allowing already infeasible packets to remain in the system would consume queue capacity and PRB-driven service time without improving timely completion. The proposed controller instead uses deadline-aware filtering to preserve sub-millisecond completed-packet tail behavior.

7.5. eMBB delivered throughput

Fig. 5 reports the instantaneous aggregate eMBB delivered throughput across both gNBs. Under nominal operation, the eMBB slice sustains approximately 12–15 Mb/s, while short peaks occur when residual PRB capacity becomes available. These peaks should be interpreted as opportunistic use of spare slice capacity rather than as a guaranteed minimum eMBB rate.

The throughput trace shows that URLLC protection does not collapse eMBB service under the representative load. The controller reserves additional PRBs for URLLC when tail-delay risk is detected, but releases capacity conservatively after recovery. As a result, eMBB continues to exploit residual capacity during URLLC underutilization periods.

7.6. Scenario-based stress evaluation

To assess whether the proposed controller remains effective beyond the representative 100-PRB setting, we additionally evaluate the stress profiles summarized in Table 7. These profiles correspond to distinct operating regimes rather than isolated single-parameter changes. Each profile selects a scenario type, global load scale, slice-specific load scale, PRB budget, burst multiplier, and new-user injection intensity. As a result, the profiles jointly change offered load, burstiness, user-arrival pressure, and available radio capacity.

In the implementation, the offered aggregate slice traffic is generated as

$$\begin{aligned}\lambda_U^{\text{off}} &= 360 \cdot L \cdot \xi_U,\\ \lambda_E^{\text{off}} &= 1040 \cdot L \cdot \xi_E,\\ \lambda_S^{\text{off}} &= 240 \cdot L \cdot \xi_S,\end{aligned} \tag{67}$$

where L is the global load-scaling factor and $\boldsymbol{\xi} = (\xi_U, \xi_E, \xi_S)$ is the slice-specific load-scaling vector. The resulting rates are aggregate offered packet rates over the simulated network.

Table 8 presents the corresponding stress test results for the proposed deadline-aware controller. The values are averaged over five independent seeds. Across all three stress profiles, the controller keeps URLLC p_{99} below the 1 ms slice-level queueing target, with values between 0.878 and 0.879 ms. The URLLC finite-horizon timely-completion ratio remains approximately 99.36%–99.38%, and the effective URLLC queue loss remains close to 0.62%–0.63%.

The stress results show that the proposed controller prioritizes URLLC tail stability under severe operating regimes. In the URLLC-weighted constrained profile, both the global load and the URLLC-specific load scale are increased. Specifically, $L = 1.7$, $\boldsymbol{\xi} = [2.4, 1.0, 1.0]$, and C_{tot} is reduced to 60, yielding an aggregate offered URLLC load of 1468.8 pkt/s. Thus, this profile represents a URLLC-dominant and resource-constrained regime rather than a simple URLLC-only perturbation. Even under this profile, URLLC p_{99} remains below 1 ms and $P_{\text{eff},U}$ remains close to 0.63%.

In the Safety-weighted constrained profile, $L = 1.6$, $\boldsymbol{\xi} = [1.0, 1.0, 2.4]$, and $C_{\text{tot}} = 60$, yielding an aggregate offered Safety load of 921.6 pkt/s. The main effect appears as a larger unfinished Safety backlog at the end of the finite observation horizon, rather than as immediate queue-level loss. In the eMBB-weighted constrained profile, $L = 1.5$, $\boldsymbol{\xi} = [1.0, 2.2, 1.0]$, and $C_{\text{tot}} = 70$, yielding an aggregate offered eMBB load of 3432.0 pkt/s. In this regime, eMBB throughput increases to approximately 39.3 Mb/s, but the finite-horizon eMBB timely-completion ratio decreases because a larger fraction of eMBB packets remains unfinished at the 180 s stress horizon.

These stress results should be interpreted as evidence of the operating region rather than universal performance guarantees for all slices under overload. They show the trade-off induced by a slicing policy that prioritizes URLLC: the controller preserves the URLLC tail-latency target, while non-URLLC slices may accumulate unfinished backlog when the dominant stress source and PRB constraint exceed the available service headroom. This distinction is important because the observed non-URLLC degradation appears mainly through P_{pend}, i.e., packets still pending at the end of the finite observation horizon, rather than through immediate queue-level loss P_{eff}.

Table 7
Stress profiles used for robustness evaluation. The profiles jointly modify the global load scale L, slice-specific load scale ξ, burst multiplier, new-user injection intensity, and PRB budget. Offered rates are aggregate packet rates generated according to (67).

Profile	Operating regime	L	$\xi = (U, E, S)$	Offered load (U, E, S) pkt/s	Burst mult. / new UEs	C_{tot}
URLLC-dominant	URLLC-weighted constrained	1.7	[2.4, 1.0, 1.0]	(1468.8, 1768.0, 408.0)	2.5 / 8	60
Safety-dominant	Safety-weighted constrained	1.6	[1.0, 1.0, 2.4]	(576.0, 1664.0, 921.6)	2.5 / 8	60
eMBB-dominant	eMBB-weighted constrained	1.5	[1.0, 2.2, 1.0]	(540.0, 3432.0, 360.0)	2.0 / 8	70

Table 8
Stress-test results for the proposed deadline-aware controller. The evaluated profiles are URLLC-dominant, Safety-dominant, and eMBB-dominant. R_{e2e} denotes the finite-horizon timely-completion ratio, P_{eff} the effective queue-loss fraction, and P_{pend} the unfinished-by-horizon backlog fraction. eMBB throughput is the aggregate delivered throughput.

Profile	URLLC				eMBB		Safety	
	p_{99} (ms)	R_{e2e} (%)	P_{eff} (%)	P_{pend} (%)	Mbps	R_{e2e} (%)	R_{e2e} (%)	P_{pend} (%)
URLLC-dominant	0.879	99.361	0.631	0.007	22.87	92.412	92.000	7.998
Safety-dominant	0.879	99.379	0.620	0.001	21.35	91.584	89.411	10.554
eMBB-dominant	0.878	99.368	0.632	0.000	39.29	83.913	91.999	7.999

Table 9
Mechanism ablation of predictive deadline dropping and tail-guided PRB control under the URLLC-activation regime.

Variant	p_{99} [ms]	R_{e2e} [%]	P_{eff} [%]	P_{pend} [%]	eMBB [Mb/s]
Proposed	0.879	99.121	0.628	0.251	22.52
No-Tail-Guidance	0.878	98.295	0.621	1.084	27.30
No-Predictive-Drop	0.942	99.105	0.000	0.268	22.64
No-Tail/No-Predictive	0.941	98.271	0.000	1.111	27.17

7.7. Mechanism ablation of predictive deadline dropping and tail-guided PRB control

To isolate the main mechanisms of the proposed controller, we perform a focused ablation study under the same traffic, mobility, channel, and queueing conditions. The results are averaged over five independent seeds. Four controller variants are compared: the full Proposed method, *No-Tail-Guidance*, in which tail-guided PRB adaptation is disabled, *No-Predictive-Drop*, in which predictive deadline dropping is disabled, and *No-Tail/No-Predictive*, in which both mechanisms are disabled. This comparison separates the effect of tail-responsive service-capacity adaptation from the effect of deadline-aware queue filtering.

Table 9 shows that the two mechanisms have distinct roles. Predictive deadline dropping is the main driver of URLLC tail protection. When it is disabled, the URLLC p_{99} delay increases from about 0.879 ms to about 0.942 ms, even though the effective queue-loss fraction drops to zero. This indicates that packets that are unlikely to complete on time remain in the system and inflate the completed-packet tail delay rather than being filtered before service.

By contrast, disabling tail-guided PRB adaptation leaves the URLLC p_{99} nearly unchanged in the considered activation regime, but it significantly worsens finite-horizon timely completion and unfinished backlog. In particular, $R_{\text{e2e},U}$ decreases from 99.121% to 98.295%, while $P_{\text{pend},U}$ increases from 0.251% to 1.084%. At the same time, eMBB throughput rises from 22.52 to 27.30 Mb/s, indicating that the full controller actively redirects service capacity toward URLLC during activation.

When both mechanisms are disabled, the system shows the weakest joint behavior: the URLLC tail remains elevated at 0.941 ms, the finite-horizon timely-completion ratio falls to 98.271%, and the unfinished backlog reaches 1.111%. The ablation therefore refines the interpretation of the proposed controller. Predictive deadline dropping primarily protects the URLLC completed-packet tail, whereas tail-guided PRB adaptation primarily improves finite-horizon timely completion and backlog control by reallocating service capacity during transient URLLC stress. Their combined action yields the most balanced URLLC protection in the proposed transient deadline-aware slicing framework.

7.8. Safety PC5/HARQ delivery ablation

To isolate the contribution of the Safety-side PC5/HARQ delivery stage, we additionally perform a five-seed ablation under the Safety-dominant regime. The comparison disables and enables HARQ while keeping the same queueing, traffic, mobility, and PRB-control settings. The metrics reported in this subsection are PC5-stage metrics and should not be confused with the queue-side finite-horizon R_{e2e} metric used in Table 6.

Table 10 shows that the first-attempt PC5 delivery reliability is already very high, approximately 99.87%. Therefore, the absolute HARQ gain is small but measurable. With HARQ enabled, the downstream PC5 delivery reliability increases from 99.8697% after the first attempt to 99.9121% after HARQ, corresponding to a gain of 0.0423 percentage points. The reported residual PC5 failure is reduced from 0.0422% in the HARQ-off case to 0.0000% in the HARQ-on case, while the mean number of HARQ attempts remains close to one. This indicates that retransmissions are rarely needed in the considered Safety-dominant regime, but they remove the small residual PC5 delivery-failure component.

The ablation clarifies the role of HARQ in the proposed Safety model. HARQ is not the dominant factor behind the Safety finite-horizon R_{e2e} gap in this stress regime. Instead, the dominant component is queue-side unfinished backlog at the observation horizon, represented by $P_{\text{pend},S}$. HARQ contributes at the downstream PC5 stage by closing the small residual PC5 delivery-failure probability. Therefore, the Safety results should be interpreted as two coupled but distinct effects: queue-side finite-horizon completion behavior and downstream PC5/HARQ delivery reliability.

7.9. Impact of service-time variance (Erlang-k_U) on URLLC tail behavior

Table 11 reports the impact of the URLLC Erlang shape parameter k_U on tail latency and finite-horizon timely-completion behavior in the representative moderate-load scenario. Increasing k_U reduces service-time variance while keeping the mean service rate fixed. The results show a clear monotonic improvement in URLLC performance as k_U increases. In particular, the URLLC p_{99} delay decreases from 0.954 ms at $k_U = 1$ to 0.747 ms at $k_U = 8$, while the effective queue-loss fraction P_{eff} decreases from 6.884% to 0.034%. Over the same range, the queue-side finite-horizon timely-completion ratio R_{e2e} increases from 93.116% to 99.965%.

A second observation is that the largest practical gain occurs in the transition from exponential or highly variable service ($k_U = 1, 2$) to the

Table 10
Safety PC5/HARQ ablation under the Safety-dominant regime. $R^{(1)}_{\mathrm{PC5}}$ denotes first-attempt PC5 delivery reliability, $R^{\mathrm{HARQ}}_{\mathrm{PC5}}$ denotes PC5 delivery reliability after HARQ, and Δ_{HARQ} is the HARQ gain in percentage points. Queue-side Safety metrics are reported separately to show that the dominant Safety finite-horizon gap is pending backlog rather than PC5 failure.

Case	$R^{(1)}_{\mathrm{PC5}}$ (%)	$R^{\mathrm{HARQ}}_{\mathrm{PC5}}$ (%)	Δ_{HARQ} (pp)	PC5 failure (%)	Mean HARQ attempts	Safety queueing $p_{99,S}$ (ms)	$R_{\mathrm{e2e},S}$ (%)	$P_{\mathrm{eff},S}$ (%)	$P_{\mathrm{pend},S}$ (%)
HARQ-off	99.8708	99.8708	0.0000	0.0422	1.0000	5.1664	89.3208	0.0409	10.6384
HARQ-on	99.8697	99.9121	+0.0423	0.0000	1.0004	5.1391	89.4108	0.0357	10.5535

Table 11
Impact of the URLLC Erlang shape parameter k_U on URLLC tail latency and finite-horizon timely-completion behavior in the representative moderate-load sensitivity sweep. Results are averaged over five seeds; P_{eff} denotes effective queue loss.

k_U	p_{99} [ms]	P_{eff} [%]	R_{cond} [%]	R_{e2e} [%]	Tail [%]
1	0.954	6.884	100.000	93.116	6.884
2	0.939	3.007	100.000	96.993	3.007
4	0.877	0.614	100.000	99.386	0.614
6	0.808	0.131	100.000	99.869	0.131
8	0.747	0.034	100.000	99.965	0.035

moderate variance-shaped regime around $k_U = 4$. At $k_U = 4$, the proposed controller already achieves $p_{99} = 0.877$ ms, $P_{\mathrm{eff}} = 0.614\%$, and $R_{\mathrm{e2e}} = 99.386\%$, while $R_{\mathrm{cond}} = 100\%$ across the full sweep. Larger values such as $k_U = 6$ and $k_U = 8$ provide additional improvement, but with diminishing marginal benefit and a progressively less variable service abstraction. For this reason, $k_U = 4$ is adopted in the representative configuration as a conservative variance-shaped setting that materially improves URLLC tail stability without assuming near-deterministic service.

8. Evaluation across multiple slicing baselines

We evaluate the proposed slicing framework against a set of representative *slicing baselines* that reflect commonly adopted control strategies for NR-V2X and URLLC-eMBB coexistence. Each baseline induces a distinct *slice-level queueing behavior*, depending on whether service capacity is static or adaptive, whether priority mechanisms are employed, and whether deadlines or feasibility constraints are enforced.

All baselines are evaluated under identical traffic loads, mobility patterns, channel realizations, and PHY/MAC abstractions. Differences in performance arise solely from the inter-slice control logic and the resulting queue dynamics.

- **Static slicing.** PRBs are partitioned according to a fixed slice split that remains unchanged throughout the simulation. As a result, the service capacity and service parallelism of each slice do not react to queue state, tail-delay risk, or channel-driven congestion. The baseline retains finite buffers, but predictive deadline dropping and other deadline-aware queue-filtering mechanisms are disabled. This setting represents semi-static slice dimensioning in which radio resources are reserved according to a fixed partition rather than adapted to transient packet-level conditions. It is conceptually aligned with fixed-capacity finite-buffer queueing abstractions [21,22] and with the general resource-partitioning view of network slicing [6,7,33].
- **Adaptive without deadline awareness.** PRBs are dynamically reallocated across slices based on congestion indicators, so the effective service capacity and service parallelism of each slice vary over time. However, predictive deadline dropping, deadline-aware queue filtering, and explicit URLLC tail-guided control are disabled. This baseline therefore captures adaptive RAN slicing that responds to load variation but does not explicitly enforce deterministic packet deadlines or control URLLC tail latency [33,34,37,40].
- **Priority-based URLLC slicing.** URLLC traffic is granted strict priority through preemptive PRB allocation, allowing URLLC packets to obtain resources by temporarily reducing the PRB share of non-URLLC slices. This reduces URLLC waiting pressure through priority access, but predictive deadline dropping, deadline-aware queue filtering, and tail-delay feedback remain disabled. From a queueing perspective, this regime is consistent with a preemptive-priority service abstraction. It represents priority-based and preemptive URLLC multiplexing mechanisms studied in the low-latency coexistence literature [17,38,51].
- **MaxWeight.** PRBs are allocated according to a MaxWeight policy that maximizes a backlog-weighted instantaneous service objective across slices. Allocation decisions depend on instantaneous queue backlogs and achievable service rates rather than deterministic deadlines, predictive deadline handling, or explicit tail-latency objectives. This baseline is therefore stability- and throughput-oriented, but it is not designed to control packet-level deadline feasibility or URLLC p_{99} delay. It represents the classical backlog-driven scheduling view established in MaxWeight theory for constrained wireless systems [45].
- **Proposed (Deadline-Aware).** The proposed framework combines dynamic PRB allocation, deterministic deadlines, predictive deadline dropping, and Gamma/Erlang service with variance shaping ($k_U = 4$). Slice queues operate as transient, deadline-aware systems of class $M(t)/G_{k_s}/c_s(t)/K_s(t)$, where PRB allocation changes packet-level service parallelism through $c_s(t)$. Unlike the above baselines, the proposed method explicitly couples tail-guided PRB control with deadline-aware queue handling, finite-horizon timely-completion evaluation, and loss decomposition.

Interpretation of effective queue loss and timely-completion metrics. For the Static, Adaptive, Priority, and MaxWeight baselines, predictive deadline dropping and related deadline-aware queue filtering are disabled. Consequently, P_{eff} in these baselines mainly reflects finite-capacity admission loss. In the present results, P_{eff} is exactly zero for the Adaptive and Priority baselines and remains negligible for the Static and MaxWeight baselines (0.00024% and 0.00133%, respectively). These near-zero values should therefore not be interpreted as successful URLLC operation, because packets may still remain in the system long enough to miss the 1 ms deadline and reduce finite-horizon timely completion. In the Proposed method, P_{eff} also includes deadline-induced queue loss generated by predictive deadline dropping. This controlled queue loss prevents packets that are unlikely to complete on time from consuming service capacity and inflating the completed-packet p_{99} delay. Therefore, the main comparison is not admission loss alone, but the joint behavior of p_{99}, R_{e2e}, P_{eff}, and Tail.

Tables 12 and 13 show that slicing strategies without explicit deadline awareness fail to control URLLC tail latency even when their effective queue loss is zero or negligible. In the Static, Adaptive, Priority, and MaxWeight baselines, packets are admitted whenever finite-buffer capacity is available, regardless of whether they can still meet the 1 ms latency target. Consequently, P_{eff} remains zero for the Adaptive and Priority baselines and is only marginally above zero for the Static and MaxWeight baselines, while the URLLC p_{99} delay still exceeds the target and the finite-horizon non-timely-completion fraction remains substantial. In particular, the Static baseline reaches $p_{99} = 2.471$ ms with a Tail value of 22.793%, Adaptive gives $p_{99} = 1.899$ ms with Tail = 9.439%,

Table 12
URLLC SLA summary across slicing baselines (target $p_{99} \leq 1$ ms). P_{eff} denotes effective queue loss, R_{e2e} denotes the finite-horizon timely-completion ratio, and Tail denotes $1 - R_{\text{e2e}}$. Results are averaged over 20 seeds.

Baseline	p_{99} [ms]	P_{eff} [%]	R_{cond} [%]	R_{e2e} [%]	Tail [%]
Static	2.471	0.00024	84.533	77.207	22.793
Adaptive	1.899	0.00000	91.149	90.561	9.439
Priority-based	1.338	0.00000	95.823	95.319	4.681
MaxWeight	1.967	0.00133	90.387	84.837	15.163
Proposed	0.879	0.62978	100.000	99.211	0.789

Table 13
URLLC p_{99} delay, effective queue loss, and eMBB delivered throughput with 95% confidence intervals. Results are averaged over 20 seeds.

Baseline	URLLC p_{99} [ms]	P_{eff} [%]	eMBB Thr. [Mb/s]
Static	2.471 [2.466, 2.475]	0.00024 [0.00006, 0.00042]	14.724 [14.608, 14.839]
Adaptive	1.899 [1.896, 1.903]	0.00000 [0.00000, 0.00000]	14.569 [14.464, 14.675]
Priority-based	1.338 [1.336, 1.340]	0.00000 [0.00000, 0.00000]	14.455 [14.315, 14.595]
MaxWeight	1.967 [1.963, 1.971]	0.00133 [0.00090, 0.00175]	14.616 [14.509, 14.723]
Proposed	0.879 [0.878, 0.879]	0.62978 [0.62245, 0.63712]	13.589 [13.460, 13.718]

Priority gives $p_{99} = 1.338$ ms with Tail = 4.681%, and MaxWeight gives $p_{99} = 1.967$ ms with Tail = 15.163%. This demonstrates that admission-loss metrics alone can be misleading for ultra-low-latency services.

Priority-based coexistence provides the best URLLC performance among the non-deadline-aware baselines, improving both URLLC delay and finite-horizon timely completion, but it still does not meet the 1 ms p_{99} target. MaxWeight maintains broadband throughput comparable to the other non-deadline-aware baselines, but its backlog-driven decisions remain inadequate for URLLC tail-latency control and lead to a markedly lower R_{e2e}. Among the evaluated slicing strategies, only the Proposed method explicitly combines deadline-aware admission, feasibility-based dropping, tail-guided PRB adaptation, and variance-shaped service. As a result, it achieves $p_{99} < 1$ ms with $p_{99} = 0.879$ ms, $R_{\text{e2e}} = 99.211\%$, and controlled effective queue loss $P_{\text{eff}} = 0.630\%$, while maintaining eMBB delivered throughput at approximately 13.59 Mb/s.

9. Discussion

The results show that deadline-aware NR-V2X slicing should be interpreted through both delay and packet completion/loss metrics. In transient operation, short queue build-up periods can determine the URLLC tail even when the average delay or the nominal PRB split appears acceptable. Therefore, the reported p_{99}, finite-horizon timely completion, queue-loss decomposition, pending backlog, and separate Safety PC5/HARQ metrics are complementary rather than interchangeable.

The improvement observed in the proposed controller is best understood as the combined effect of three mechanisms. PRB reallocation changes the number of packets that can be served in parallel, deadline-aware queue handling prevents already infeasible packets from consuming service capacity, and tail-guided feedback reacts to short-term URLLC delay pressure. The stress and ablation results also show the cost of this service hierarchy: when URLLC pressure increases, non-URLLC traffic may accumulate more unfinished backlog at the finite observation horizon. This behavior reflects a URLLC-prioritizing slicing policy in which the latency-critical slice is protected first, while eMBB and Safety effects remain visible through separate backlog and delivery metrics.

The simulation design clarifies the aspects of the V2X system being tested. The waypoint mobility model creates time-varying gNB association, path loss, fading, and interference conditions, while the time-varying Poisson and burst processes generate controlled transient load variations. These choices allow the slice-control mechanisms to be tested under repeatable conditions while independently changing the load level, burst intensity, and slice-specific stress. Therefore, they are useful for isolating the effect of deadline-aware PRB control. However, in deployment-oriented V2X studies, vehicle motion and message generation are more tightly coupled. Lane-level car-following behavior, road topology, intersections, traffic-light phases, and congestion patterns can change both vehicle density and the temporal structure of V2X messages. Such effects are commonly studied using microscopic traffic simulations, for example, with SUMO [63], coupled with V2X simulation frameworks such as Veins [64] or VaN3Twin [65].

This distinction is important for the traffic model. Periodic Cooperative Awareness Messages (CAMs) are not always generated at a fixed interval; their generation can depend on vehicle dynamics and may vary over a range, such as 100 ms to 1 s [25]. Event-driven Decentralized Environmental Notification Messages (DENMs) are also coupled with local road events [26]. Under high traffic density, variable message generation can interact with semi-persistent sidelink resource reservation, where sidelink resources are reserved over repeated intervals and may become underutilized when the actual message generation rate decreases. The congestion studied in this paper therefore represents scheduler-level slice congestion under controlled transient load, whereas a full deployment study would additionally capture mobility-driven channel congestion and standards-compliant message-generation dynamics.

The choice of NR-V2X is based on the RAN slicing focus of this study. The controller acts on a gNB-side PRB budget and studies how PRB-limited service capacity can be shared among URLLC, eMBB, and Safety traffic. This makes a 5G NR-V2X setting a natural reference point because it combines infrastructure scheduling over the Uu interface, i.e., the UE-to-gNB cellular link, slice-aware RAN resource control, and PC5 sidelink delivery for safety-oriented traffic [4]. The separate PC5/HARQ abstraction is used to keep gNB queueing effects and sidelink delivery effects distinguishable, rather than to reproduce every NR-V2X PHY/MAC procedure. Detailed sidelink procedures, such as resource sensing and reselection, sidelink control signaling, configured grants, modulation-and-coding-scheme (MCS) adaptation, transport-block construction, and exact HARQ feedback timing, are represented at the scheduler interface through service-time variation, PRB requirements, fading, and BLER/HARQ abstractions. Accordingly, the framework is applicable as a scheduler-level control layer that can be coupled with standards-compliant NR-V2X implementations, while the reported gains should be interpreted as PRB-based slice-control gains rather than full-stack PHY/MAC optimization gains.

Other V2X radio-access technologies are also important. IEEE 802.11p, IEEE 802.11bd, and ETSI ITS-G5 are widely discussed for V2X deployments and pilot programs, particularly in the European context [66,67]. These technologies differ from cellular V2X in terms of their access and resource-control mechanisms, including contention-based distributed channel access in IEEE 802.11p/802.11bd/ITS-G5 and sensing- or scheduling-based sidelink resource allocation in cellular V2X systems [66,67]. Therefore, they represent complementary deployment paths rather than direct substitutes for the gNB-side PRB-slicing problem studied here. A direct comparison would require technology-specific models for contention, channel load, access delay, and service capacity.

A deployment-oriented extension would connect the proposed queueing and PRB-control logic to realistic mobility traces and standards-compliant V2X traffic generation. SUMO-based mobility, CAM/DENM generation rules, and V2X simulation frameworks, such as Veins or VaN3Twin, can be used to test whether the same deadline-aware control trends persist under lane-level mobility, intersections, congestion waves, variable message periodicity, and semi-persistent resource reservation.

10. Conclusion

This study shows that controlling the URLLC tail latency in vehicular NR-V2X systems requires more than static slicing, backlog-driven adaptation, strict priority, or stability-oriented scheduling alone. Under mobility-driven traffic variation and time-varying service capacity, these strategies do not provide the same level of control over high-percentile URLLC delay.

To address this problem, we developed a transient, deadline-aware RAN slicing framework based on $M(t)/G_{k_s}/c_s(t)/K_s(t)$ slice queues, where PRB allocation directly determines the packet-level service parallelism. The proposed method combines transient finite-capacity queue modeling, predictive deadline dropping, variance-shaped URLLC service, and tail-guided PRB adaptation within a causal discrete-event simulation framework. In the representative NR-V2X scenario, the proposed controller maintains URLLC p_{99} between 0.85 and 0.92 ms across the two gNBs and achieves a 99.37% finite-horizon URLLC timely-completion ratio with 0.63% effective URLLC queue loss.

The results also show that URLLC tail protection can be achieved without sacrificing broadband service. While prioritizing URLLC under transient stress, the controller continues to support eMBB throughput and uses available radio resources efficiently across slices. Taken together, the results support the use of transient queue modeling, predictive deadline-aware queue handling, and tail-guided PRB control for NR-V2X RAN slicing under time-varying vehicular conditions.

From an implementation perspective, the controller relies on short-horizon queue and delay measurements, uses lightweight local control logic, and remains suitable for near-real-time slice management at the gNB level. This makes the framework practically relevant for deadline-sensitive multi-slice vehicular RAN operations.

11. Future work

The proposed framework can be extended in several ways.

- **Analytical approximations.** Developing fluid or diffusion approximations for transient deadline-aware $M(t)/G_{k_s}/c_s(t)/K_s(t)$ queues would provide additional analytical insights into tail-delay and backlog behavior.
- **Learning-enhanced control.** Learning-based or model-predictive extensions can be combined with the current tail-guided controller to improve adaptation under rapidly changing vehicular traffic and radio conditions.
- **Sidelink-aware extensions.** A tighter integration of gNB-side slicing with PC5 sidelink resource sensing, resource reselection, and 3GPP-assisted sidelink scheduling would allow the framework to capture richer interactions between queueing, PRB control, and vehicular safety delivery.
- **Trace-based and deployment-oriented validation.** Future validation can use SUMO-based mobility traces, standards-compliant CAM/DENM traffic generation, and V2X simulation frameworks such as Veins or VaN3Twin, together with experimental platforms such as OAI, 5G/ICE, or 5G/6G testbeds. This would allow the controller to be evaluated under lane-level mobility, density-dependent message generation, variable CAM periodicity, semi-persistent resource reservation, and realistic channel-congestion dynamics.
- **O-RAN implementation.** Embedding the tail delay and queue-state control logic in an O-RAN Near-RT RIC xApp or in a slice-aware MAC scheduler would be a practical step toward deployment-oriented evaluation.
- **Uplink and multi-cell extensions.** Extending the framework to uplink scheduling, inter-cell coordination, and richer interference coupling would provide a more complete view of NR-V2X slice behavior.
- **Hybrid Uu/PC5 operation.** Future studies can also investigate hybrid delivery strategies in which latency-critical traffic dynamically uses gNB forwarding, PC5 sidelink, or relay-assisted transmission according to the current queue and channel conditions.

These directions show how the proposed transient queueing and control framework can be developed further for broader analytical, algorithmic, and deployment-oriented studies on vehicular RAN slicing.

CRediT authorship contribution statement

Nurgüneş Yordanov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing; **Bülent Çavuşoğlu:** Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code availability

The simulation code used in this study is available from the corresponding author upon reasonable request.

Data availability

Data will be made available on request.

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